

Uncovering Culture-Induced Gene Expression Changes in HGSOC Organoids

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1. Abstract

High-grade serous ovarian carcinoma (HGSOC), the deadliest and most common form of ovarian cancer, with an estimated 600,000 women living with the disease, globally. Patient-derived 3D organoids (PDOs) are premier translational models. However, the transcriptomic consequences of transitioning malignant cells from nutrient-stressed *in vivo* effusion conditions to nutrient-rich *in vitro* culture remain poorly characterized. If transitioning to the lab radically alters gene expression immediately, drug screening and biological conclusions can be skewed. Therefore, characterizing these early changes is vital to guarantee laboratory data mirrors real patient outcomes. Characterization ensures that culture-induced effects are isolated from patient-specific variations. RNA-seq analysis was performed on HGSOC PDOs from four patients at Day 0 (baseline) and Day 6 post-isolation (8 total samples; 58,725 genes). Paired differential gene expression analysis was conducted using DESeq2, modeling patient-specific baselines alongside temporal culture adaptation. Transitioning from *in vivo* conditions to *in vitro* conditions triggered metabolic reprogramming, and subsequent transcriptomic shift. Genes associated with hypoxia responses, inflammation, apoptosis, and other immediate stress response (such as *FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8 (IL-8)*, *MGMT*, and *SLC11A1*), were significantly downregulated. Conversely, genes associated with cell proliferation, DNA replication, biomass production and cell cycle progression (such as *AURKA (Aurora Kinase A)*, *PTTG1 (Securin)*, *FANCI* and *NPAT*). Crucially, Principal Component Analysis (PCA) revealed that patient-specific genetic heterogeneity remained the dominant driver of global variance (PC1: 38%), overshadowing culture-induced adaptation (PC2: 24%). Characterizing this transcriptomic shift is vital because it proves that the *in vitro* organoid models are adapting, as opposed to mutating. It confirms that while the organoids successfully stabilize *in vitro*, they still preserve the patient's unique tumor architecture and biology (i.e., genetic, transcriptomic, and molecular profile). This study proves that HGSOC organoids undergo predictable adaptation rather than random genetic divergence, and validates their integrity as premier functional platforms for high-throughput drug screening and precision oncology. HGSOC organoids remain reliable patient-specific models, despite the massive transcriptomic shift.

Keywords: Patient-derived organoids (PDOs), High-grade serous ovarian carcinoma (HGSOC), *in vitro* adaptation, tumor microenvironment (TME), metabolic reprogramming, differential gene expression, DESeq2, RNA-sequencing (RNA-seq), principal component analysis, stress, hypoxia, translational medicine, precision oncology.

2. Introduction & Research Problem

High-grade serous ovarian carcinoma (HGSOC), the deadliest and most common form of ovarian cancer, is plagued by chemotherapy resistance, making physiologically relevant models crucial for clinical research (Tadić et al., 2024). Globally, ~ 600,000 women live with HGSOC ² (Ren et al. 2025). Historically, if researchers wanted to test a cancer drug in a lab, they grew tumor cells flat on a plastic dish. However, cells don't live on flat plastic inside the body; they live in a complex, 3D world. Flat cells behave differently and don't preserve the true patient tumor architecture and biology, hence are not useful in drug screening.

PDOs are the ultimate functional bridge between translational medicine and precision oncology. An organoid is a tiny, 3D "mini-tumor" or "mini-organ" grown in a dish from a patient's actual cells (Figure 1). PDOs are premier tools and vastly outperform traditional 2D cultures by preserving true patient tumor architecture and biology (Tadić et al., 2024) (Qi et al., 2025). By cultivating a patient's own tumor cells into 3D organoids, researchers can preserve the exact genetic, transcriptomic, and molecular heterogeneity of the patient's unique disease. Ultimately, these 3D organoids are then used to study cancer progression and test personalized chemotherapy treatments.

In the context of translational medicine, these 3D models allow researchers to bridge the gap from drug discoveries in the lab to administering them to actual patients (after clinical trials and rigorous regulatory approval), generating insightful and biologically-relevant data. This allows clinicians to screen hundreds of therapeutic targets *in vitro* to predict how a patient's actual tumor will respond to treatment in theory, before in practice.

2.1. Importance of characterizing these early changes

If transitioning to the lab radically alters gene expression immediately, drug screening and biological conclusions can be skewed. Therefore, characterizing these early transcriptomic changes is vital to guarantee laboratory data mirrors real patient outcomes. Characterization ensures that culture-induced effects are isolated from patient-specific variations.

Characterization is important because:

- i. **It guarantees that the organoid is a true representative of the patient's tumor.**

To find the right cancer drug, scientists can grow a 3D organoid (i.e., a replica of a patient's tumor) in a dish and test different treatments on it first. When you take cells out of the body (*in vivo*) and put them in a plastic lab dish, they have to adapt to the new *in vitro* conditions. If that transition completely changes the transcriptomic profile, the HGSOC organoid model loses its patient-specific identity, and is no longer a copy of that specific patient's cancer. Therefore, the organoid would no longer be useful for testing personalized drugs suitable for that specific patient.

ii. It prevents false positives during drug screening

Inside a patient's body, cancer cells are often starving for nutrients. Once they are transferred to nutrient-rich *in vitro* conditions, the cells enter an actively proliferative state and start multiplying incredibly fast. If the potential chemotherapy drug is tested on these Day 6 organoids (i.e., the fast-growing lab cells), and is successful, it is important to decipher whether the drug worked because it's a perfect match for that patient's unique cancer, or whether the cells' temporary rapid growth is just induced by the nutrient-rich lab media, making the drug look better than it actually is. Characterization ensures that temporary, culture-induced metabolic by-product are not mistake for a genuine therapeutic target. Otherwise organoid model could lose its patient-specific molecular identity, rendering subsequent drug screens inaccurate or clinically irrelevant

Furthermore, differentiating temporary, culture-induced gene expression changes from a patient's true tumor biology is essential in preventing false positives and guaranteeing that laboratory data reflects genuine patient outcomes (i.e., that if the drug successfully kills the cancer cells in the lab (*in vitro*), it has the highest possibility of actually working inside a specific patient's body (*in vivo*).

iii. It increases the statistical power.

Human genetic diversity is massive. Every human being has a completely unique genetic blueprint. The baseline differences between Patient A and Patient B are vastly larger than the subtle transcriptomic changes that occur between Day 0 and Day 6. Running a standard, unpaired analysis would cause the huge genetic "noise" of patient-to-patient variation to completely drown out the "signal" of their 6-day adaptation. Therefore, by using a "paired" statistical approach, which essentially compares each patient's lab cells *only* back to their own original cells, the background "genetic noise" is muted, maximizing statistical power and isolating culture-specific transcriptomic adjustments. This allows researchers pinpoint the exact genes that change when cells adjust to lab conditions.

This study successfully characterized the extent to which the early-stage, short-term (Day 0 to Day 6) transcriptomic shift that occurs after the transition of patient-derived HGSOC organoids from an *in vivo* nutrient-stressed, hypoxic, malignant effusion environment to an *in vitro* nutrient-rich environment. It was hypothesized that the transfer of HGSOC cells from a nutrient-stressed, hypoxic, malignant effusion environment to an *in vitro* nutrient-rich environment will trigger a significant transcriptomic shift, relieving hypoxia signalling, downregulating genes associated with survival signalling pathways and anti-apoptosis, and upregulating pathways associated with cell proliferation.

To test this hypothesis, RNA-seq analysis was performed on HGSOC PDOs from four patients at Day 0 (baseline) and Day 6 post-isolation (8 total samples; 58,725 genes). Paired differential gene expression analysis was conducted using DESeq2, modeling patient-specific baselines alongside temporal culture adaptation.

2.2. How organoids are generated

This diagram illustrates the generation of organoids from tumor/cancer tissue. In addition to tumor/cancer tissue, organoids can also be generated from pluripotent stem cells, embryonic stem cells, and from healthy tissue samples directly from the patient's organ. In the case of this study, once the cells are isolated from the malignant effusion fluid surrounding the ovaries of the HGSOC patient, they are then grown in culture as 3D organoids on a hydrogel extracellular matrix that provides the physical scaffolding for the cells to anchor, self-organize, interact and form biochemical networks between themselves (Figure 1). Alternatively, 3D organoids can also be printed using a bioink made of living cells, supported by a nutrient-rich hydrogel that acts as the extracellular matrix (ECM).

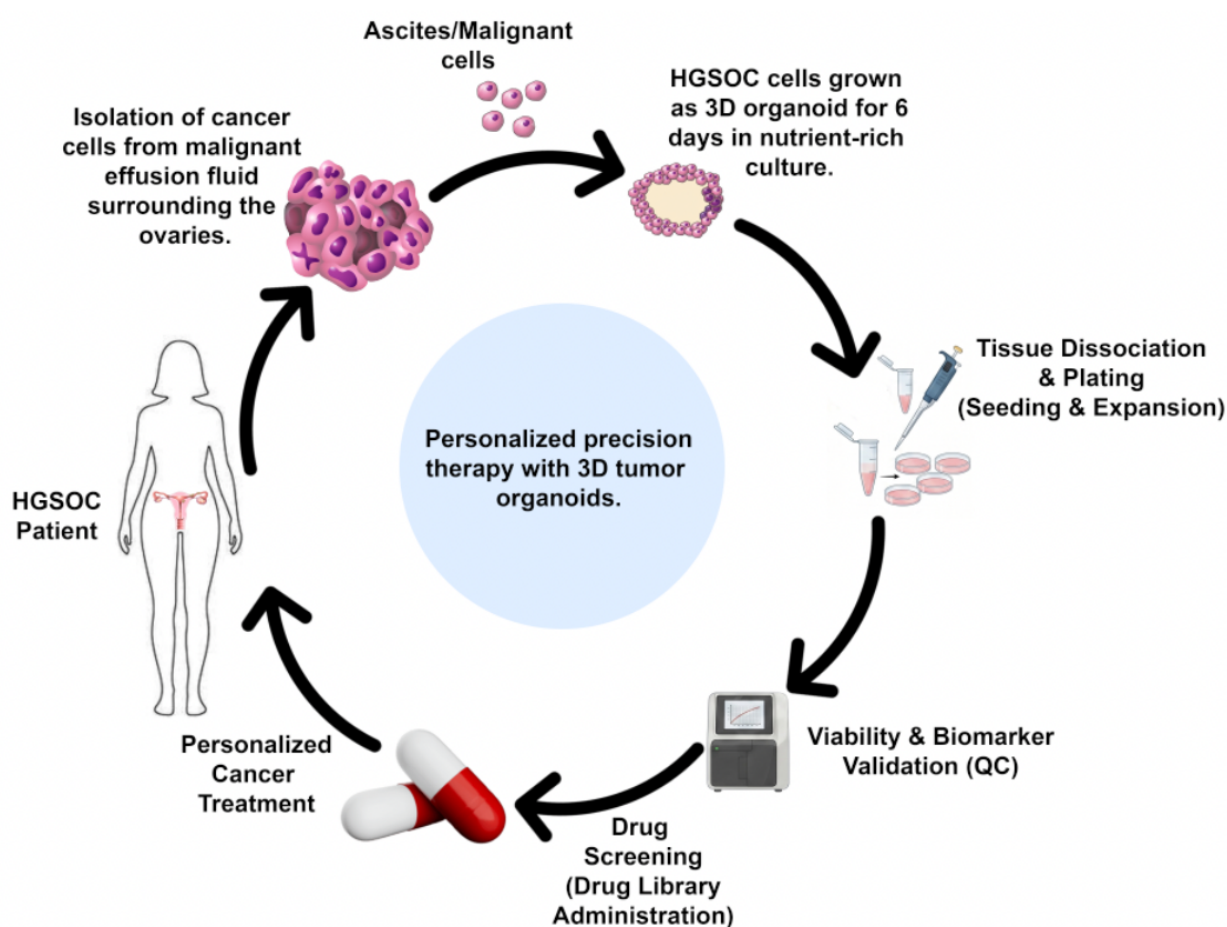


Figure 1. Schematic workflow of a patient-derived 3D organoid pipeline for functional precision medicine in HGSOC. Malignant cells are isolated from the ascites or fluid surrounding the ovaries of a High-Grade Serous Ovarian Carcinoma (HGSOC) patient. The cells are cultured as 3D tumor organoids for 6 days in a nutrient-rich medium to establish structural and genomic architecture. Following initial culture, organoids undergo tissue dissociation and are replated into multi-well plates for seeding and expansion. Automated phenotypic quality control (QC) validates cell viability and biomarker expression prior to high-throughput drug library screening. The resulting therapeutic profile translates directly into personalized cancer treatments for the patient.

1. Research Objectives & Hypothesis

1.1. Research Question

To what extent does the short-term transition of patient-derived HGSOC organoids from an *in vivo* nutrient-stressed, hypoxic, malignant effusion environment to an *in vitro* nutrient-rich environment alter their transcriptomic profile?

1.2. Hypothesis

The transfer of HGSOC cells from a nutrient-stressed, hypoxic, malignant effusion environment to an *in vitro* nutrient-rich environment will trigger a significant transcriptomic shift, relieving hypoxia signalling, downregulating genes associated with survival signalling pathways and anti-apoptosis, and upregulating pathways associated with cell proliferation.

1.3. Prediction

In the presence of new conditions (in vitro), the HGSOC cells will undergo metabolic reprogramming, the RNA-seq data on Day 6 will show clear downregulation of survival and hypoxia markers (such as *FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8 (IL-8)*, and *MGMT*) alongside systemic upregulation of cell cycle and proliferation drivers (such as *AURKA*, *PTTG1*, *FANCI* and *NPAT*) across all patient lines compared to Day 0.

1.4. Goal

To characterize the early-stage transcriptomic and biological changes that occur when patient-derived HGSOC organoids adapt to a laboratory environment over a short-term period (specifically from Day 0 to Day 6).

2. Data Set

Parameter	Description
Data type	RNA-seq gene expression count data.
Model system	Patient-derived High-Grade Serous Ovarian Carcinoma (HGSOC) organoids.
Samples	8 total RNA-seq samples (4 patients × 2 time points).
Genes measured	58,725 genes.
Source	Patient-derived organoids cultured over time.

2.1. Why Use Organoids?

HGSOC Organoids are:	Supporting Evidence
More useful model than conventional cell culture models	Patient-derived HGSOC organoids more accurately mimic patient tumors than conventional cell culture models, making them a more physiologically relevant model.
More accurate than traditional 2D cultures	Unlike 2D cultures, organoids grow in a 3D structure that preserves cell–cell and cell–matrix interactions, better mimicking the original tumor.
Important for modeling human tissue	Because human tissues are naturally 3D, organoids better replicate tissue architecture and cellular behaviour than 2D models.
Biologically meaningful tumor models	Organoids preserve tumor microenvironment and reproduce many of the molecular and phenotypic characteristics observed in patient tumors, generating more biologically meaningful results.
Overall very valuable	Patient-derived organoids closely mimic human tumors, and thus enable studies of disease mechanisms, biomarkers, transcriptomic changes, and therapies, supporting translational cancer research and precision medicine.

Table 1. Physiological relevance and advantages of patient-derived HGSOC 3D organoids. A summary of key structural and biological properties validating 3D organoids over conventional model systems. The matrix highlights how these patient-derived models maintain critical cell–cell and cell–matrix interactions, preserve the native tumor microenvironment, and mirror cellular behaviour more accurately than traditional 2D monolayers. Consequently, these biologically meaningful platforms serve as robust tools for translational research, offering a high-fidelity system for investigating disease mechanisms, transcriptomic variations, and precision oncology therapies.

2.2. What is Metabolic Reprogramming?

Metabolic reprogramming is the biological engine driving the transition (i.e., transcriptomic/gene expression shift) from **Day 0** to **Day 6**. Metabolic reprogramming is an established hallmark of cancer, where cells dynamically alter their metabolic pathways to satisfy specific demands for energy, biomass production, and survival in response to changes in their physical microenvironment (in this case, from *in vivo* to *in vitro*).

Tumor microenvironment (<i>in vivo</i>)	Non-tumor microenvironment (<i>in vitro</i>)
Day 0	Day 6
HGSOC cells experience complex, low oxygen (hypoxia), nutrient-starved conditions, stromal fibroblasts, immune cells, endothelial cells, cytokines, growth factors, and other microenvironmental stresses.	Nutrient-rich conditions (i.e., cells are relieved of microenvironmental stresses).
Upregulation of genes associated with hypoxia responses, inflammation, apoptosis, and other immediate stress response (such as <i>FOS</i> , <i>FOSB</i> , <i>JUN</i> , <i>EGR1</i> , <i>CXCL8 (IL-8)</i> , <i>MGMT</i> , and <i>SLC11A1</i>).	Downregulation of genes associated with hypoxia, inflammation and immediate stress response.
Downregulation of genes associated with cell proliferation, DNA replication, biomass production and cell division.	Upregulation of genes associated with cell proliferation, DNA replication, biomass production and cell cycle progression (such as <i>AURKA (Aurora Kinase A)</i> , <i>PTTG1 (Securin)</i> , <i>FANCI</i> and <i>NPAT</i>). HGSOC cells undergo metabolic reprogramming toward growth, increasing cell-cycle and proliferation pathways.

Table 2. Transcriptomic and environmental shifts in HGSOC cells between *in vivo* tumor and *in vitro* organoid microenvironments. A comparative matrix outlining the microenvironmental conditions and corresponding gene expression signatures at Day 0 and Day 6. At Day 0, the freshly isolated cells reflect the complex, hypoxic, and nutrient-starved native tumor microenvironment (*in vivo*), characterized by the upregulation of hypoxia, inflammatory, and immediate stress-response genes (*FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8 (IL-8)*, *MGMT* and *SLC11A1*). By Day 6, culture in nutrient-rich, *in vitro* 3D conditions relieves these microenvironmental stresses, resulting in the downregulation of acute stress pathways and a robust metabolic reprogramming toward active growth, cell cycle progression, and proliferation pathways as evidenced by the upregulation of *AURKA*, *PTTG1*, *FANCI*, and *NPAT*.

3. Experimental Design & Methodology

The biological samples for this study were derived from external workflows. Cells were isolated from the malignant effusion fluid of High-Grade Serous Ovarian Carcinoma (HGSOC) patients and grown as 3D organoids. RNA was then extracted and sequenced from these organoids at Day 0 and Day 6.

The scope of this analysis begins at the downstream bioinformatics stage. After retrieving the raw RNA sequencing data, raw read counts were first inspected and verified. Computational and statistical analysis was then initiated with a preliminary exploratory analysis and normality testing, followed by preprocessing and filtering out of low read counts.

A Differential Gene Expression Analysis (DESeq2) was then conducted, branching off into two parallel paths; statistical testing (which incorporated the Wald Test, p-value calculations, and Benjamini-Hochburg FDR correction, and a parallel quality control (QC) path using Variant Stabilizing Transformation (VST). Results of this QC path were visualized using a Global PCA Plot and a Sample Distance Heatmap.

Finally, data from both the statistical testing and QC pathways converged for the identification and interpretation of significant genes, which included the creation of a Volcano Plot of differential gene expression (Figure 6), and a heatmap highlighting the top 100 Differentially Expressed Genes (DEGs) (Figure 7).

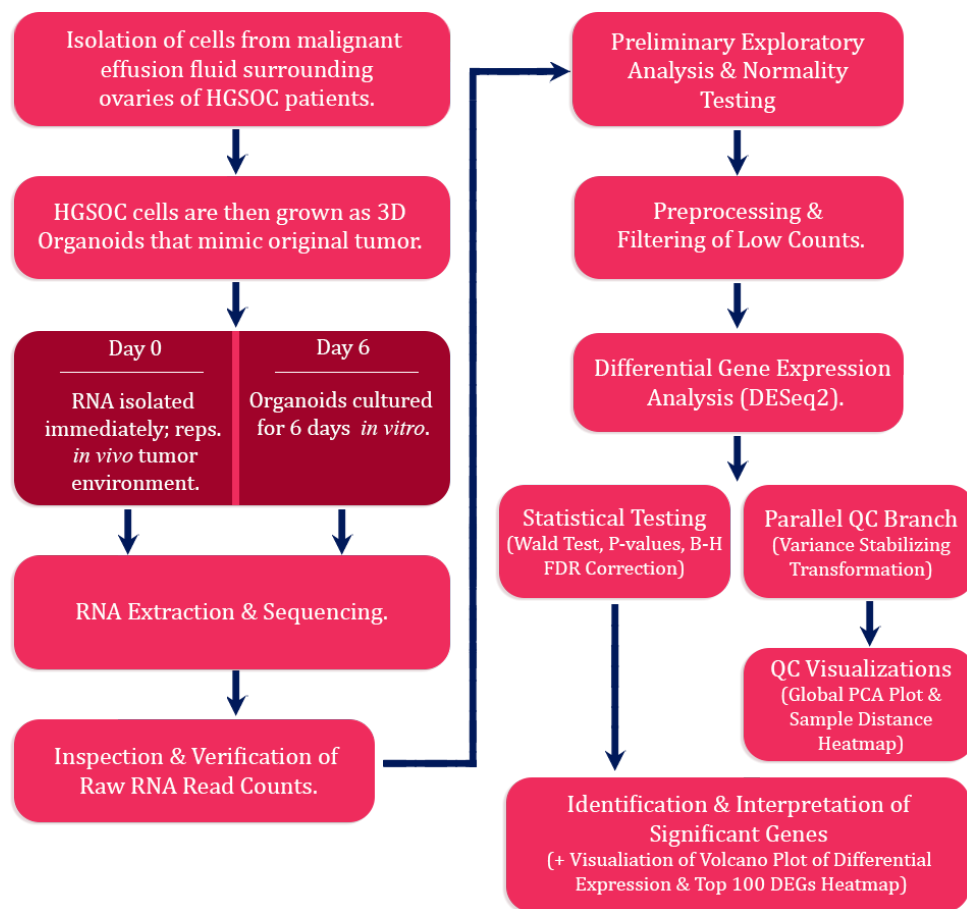
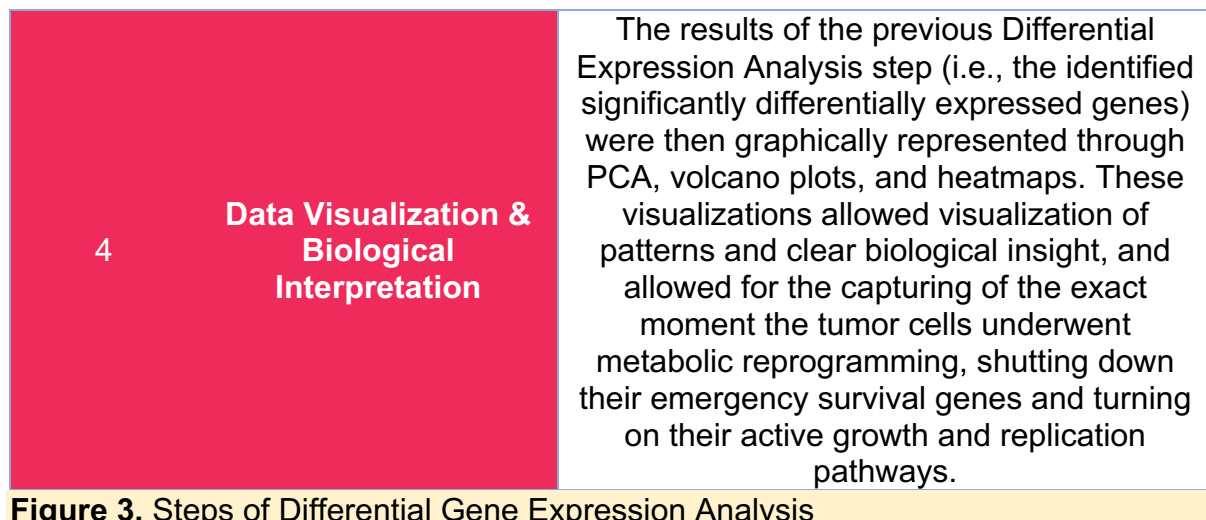


Figure 2. Experimental and bioinformatics workflow for differential gene expression analysis in HGSOc 3D organoids. Cells isolated from the malignant effusion fluid of High-Grade Serous Ovarian Carcinoma (HGSOc) patients were cultured as 3D organoids to recapitulate the native tumor architecture. Samples were processed across two temporal conditions: baseline immediate isolation (Day 0, representing the *in vivo* environment) and after a 6-day *in vitro* culture period (Day 6). Following RNA extraction and high-throughput sequencing, raw RNA read counts were inspected and verified. The downstream analytical pipeline comprised preliminary exploratory analysis, low-count filtering, and differential expression analysis via DESeq2. Statistical evaluation was performed using the Wald test with Benjamini-Hochberg False Discovery Rate (B-H FDR) correction, parallel to a quality control (QC) branch utilizing variance stabilizing transformation for Global PCA and sample distance visualization. The pipeline culminates in the identification and

functional interpretation of significant differentially expressed genes (DEGs) visualized via volcano and heatmap plots.

3.1. Steps of Differential Expression Analysis





3.1.1. Wald Test

A statistical method used to evaluate whether specific parameters in a model (such as regression coefficients) are significantly different from zero or a hypothesized value. Essentially, it determines whether removing or restricting those parameters would meaningfully degrade the model's fit.

The Wald test produces a raw p-value for each gene, which is then adjusted/corrected by the Benjamini-Hochberg (BH) FDR correction to control for false positives. Significantly differentially expressed genes are determined after this BH FDR correction.

3.1.2. Benjamini-Hochberg FDR correction

The Benjamini-Hochberg False Discovery Rate (FDR) correction is a statistical method used to control the inflation of false positives, when analyzing thousands of genes, simultaneously. Without correction, unadjusted significance threshold of $p < 0.05$ would yield thousands of false positives by random chance alone.

The BH procedure addresses this issue by adjusting p-values when performing many hypothesis tests simultaneously, such as in this analysis, where differential expression analysis was performed on thousands of genes.

Genes with an adjusted p-value (p_{adj}) below the chosen threshold ($p < 0.05$) are then identified as significantly differentially expressed genes.

3.1.3. Gene Ontology Enrichment Analysis (Day 6)

To identify enriched biological processes and pathways, Gene Ontology (GO) enrichment analysis was performed separately (at Day 6) on the significantly upregulated ($\log_2FC > 0.58$, $p_{adj} < 0.05$) and downregulated ($\log_2FC < -0.58$, $p_{adj} < 0.05$) gene sets using the Enrichr web tool (<https://maayanlab.cloud/Enrichr/>). The GO Biological Process was utilized to identify biological process associated with each gene set. Statistical significance was adjusted for multiple testing using the

Benjamini-Hochberg procedure, with enriched pathways defined at an adjusted threshold of $q < 0.05$.

4. R Analysis and Results

★ INSTALL PACKAES & LOAD THE REQUIRED LIBRARIES ★

```
``{r}
# Install Packages
# BiocManager::install("DESeq2")
# install.packages("BiocManager")
# if (!requireNamespace("pheatmap", quietly = TRUE)) install.packages("pheatmap")
# if (!requireNamespace("BiocManager", quietly = TRUE))
#   install.packages("BiocManager")
# BiocManager::install("clusterProfiler")

# Load Libraries
library(tidyverse)
library(rlang)
library(DESeq2)
library(dplyr)
library(ggplot2)
library(pheatmap)
library(clusterProfiler)
library(org.Hs.eg.db)
``
```

4.1. PRELIMINARY EXPLORATORY ANALYSIS

```
```{r}
base_path <- "./read_counts/"
output_dir <- "./"

#Merge TSV Files
files <- c(
 "A778_D0_gene.tsv", "A778_D6_gene.tsv",
 "A820_D0_gene.tsv", "A820_D6_gene.tsv",
 "A870_D0_gene.tsv", "A870_D6_gene.tsv",
 "A899_D0_gene.tsv", "A899_D6_gene.tsv"
)

#Extract sample names
sample_names <- str_remove(files, "_gene\\.tsv")

#standardize the files
list_of_dfs <- map2(files, sample_names, function(file, sample_name) {
 read_tsv(paste0(base_path, file), show_col_types = FALSE) %>%
 rename(gene_id = 1, gene_name = 2, !!sample_name :=3) %>%
 dplyr::select(gene_id, !!sample_name)
})
```

```

})

Merge dataframes into one, by gene_id
count_matrix_df <- purrr::reduce(list_of_dfs, left_join, by = "gene_id")

#Format to matrix object for DESeq2
count_matrix <- count_matrix_df %>%
 column_to_rownames("gene_id") %>%
 as.matrix()

#Convert to dataframe for saving
count_matrix %>%
 as.data.frame() %>%
 rownames_to_column(var = "gene_id") %>%
 write_csv(file = paste0(output_dir, "merged_count_matrix.csv"))

```

```

4.2. DATA INSPECTION & VERIFICATION

```

```{r}
head(count_matrix)
View(count_matrix)
dim(count_matrix) #58,725 rows x 8 columns
str(count_matrix)
colnames(count_matrix)
summary(count_matrix)
colSums(is.na(count_matrix))

```

-----
★ CHECK FOR MISSING VALUES ★
-----

```{r}
#Check for missing values (i.e., N/A's across the entire merged matrix)
total_na_matrix <- sum(is.na(count_matrix))
cat("Total missing values in the matrix:", total_na_matrix, "\n")

#Missing values per column (i.e., per sample)
na_by_sample <- colSums(is.na(count_matrix))
print("Missing values per sample:")
print(na_by_sample)

#ID rows(genes) with missing values
genes_with_na <- rownames(count_matrix)[rowSums(is.na(count_matrix)) > 0]

#Safety check to see if vector has length > 0 (i.e., if any rows have missing data)
if(length(genes_with_na) > 0) {
 cat("\nFound", length(genes_with_na), "genes with missing data. First few:\n")
}

```

```

print(head(genes_with_na))
} else {
 cat("\nClean dataset: No genes have missing (N/A) count values!\n")
}

...

```

#### ★ Interpretation of Data Inspection and Verification:

1. No genes have missing values (N/A's).
2. '0 counts' are different from N/A's.
3. 0 counts = 0 reads for a given sample after sequencing. Gene may be completely turned off or not expressed in given tumor organoid. Or the expression level of that given gene may be too low for the sequencer to detect.

### 4.3. FACETED/MERGED HISTOGRAMS FOR ALL SAMPLES

#### Log-Transformation & Visualizing Transcriptome Count Distributions

```

```{r}
#Convert matrix to long df for ggplot
matrix_long <- count_matrix %>%
  as.data.frame() %>%
  rownames_to_column(var = "gene_id") %>%
  pivot_longer(
    cols = -gene_id,
    names_to = "Sample",
    values_to = "Read_Count"
  ) %>%

#Add metadata for colouring
mutate(
  Day = if_else(str_detect(Sample, "D0"), "Day 0", "Day 6"),
  Patient = str_extract(Sample, "A[0-9]+")
)

#Plot multifaceted grid with all samples
p_faceted <- ggplot(matrix_long, aes(x = log2(Read_Count + 1), fill = Day)) +
  geom_histogram(bins = 50, colour = "white", alpha = 0.8) +
  scale_fill_manual(values = c(
    "Day 0" = "skyblue",
    "Day 6" = "lightgreen"
  )) +
  facet_wrap(~ Sample, ncol = 4) +
  labs(
    title = "Transcriptome Count Distributions Across All Samples",
    subtitle = "Comparing HGSOC Organoids at Day 0 v. Day 6",
    x = "log2(Read Count + 1)",
    y = "Number of Genes",
    fill = "Timepoint"
  ) +

```

```

theme_minimal() +
theme(
  strip.background = element_rect(fill = "grey95", color = "grey80"),
  strip.text = element_text(face = "bold"),
  panel.spacing = unit(1, "lines")
)

# Save multifaceted ggplot
ggsave(
  filename = paste0(output_dir, "Faceted_Histogram_All_Samples.png"),
  plot = p_faceted, width = 12, height = 6, dpi = 300
)

print(p_faceted)
` ``

```

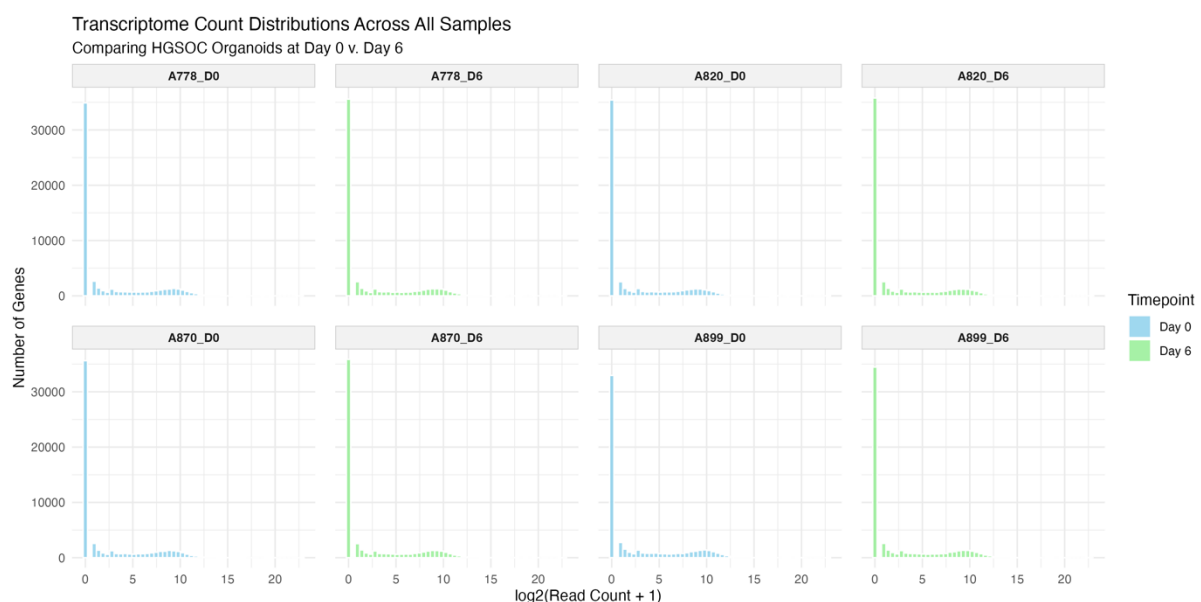


Figure 4. Transcriptome count distributions across HGSOc organoid samples at Day 0 and Day 6. Faceted histograms illustrate the distribution of gene expression levels (number of genes) as a function of $\log_2(\text{Read Count} + 1)$ for four patient-derived HGSOc organoid lines (A778, A820, A870, and A899). Samples are compared between immediate post-isolation baseline (Day 0, light blue) and after a 6-day *in vitro* culture period (Day 6, light green). All samples display highly comparable, characteristic bimodal distributions featuring a sharp, dominant peak near zero (representing unexpressed or lowly expressed genes), followed by a secondary, broader distribution representing moderately to highly expressed transcripts, confirming consistent sequencing depth and profile alignment across conditions.

4.3.1. What is Log-Transformation?

Raw RNA-seq data is highly skewed. Plotting these raw sequencing read counts on the same graph will result in the highly expressed genes will “shoot” off the graph,

while lowly-expressed genes will get “squished” into a virtually invisible line at the bottom of the graph.

Log-transformation (specifically $\log_2$ transformation) is a crucial step used to convert this highly-skewed, exponential gene expression data into manageable and statistically stable data. Log transformation essentially compresses the highly expressed genes, and “stretches” the lowly expressed genes, such that the graph becomes more symmetrical.

This symmetry allows for easier plotting, interpretation, and visual comparison of up-regulated genes (such as *AURKA*) against down-regulated genes (such as *JUN*).

Additionally, log transformation stabilizes variance (homoscedasticity), ensuring that highly expressed genes don't dominate the downstream statistical analysis purely because of their scale.

4.4. VISUALIZATION OF DISTRIBUTION OF ALL SAMPLES

```
```{r}
#Convert to long format for ggplot
long_counts <- count_matrix %>%
 as.data.frame() %>%
 rownames_to_column(var="Gene") %>%
 pivot_longer(cols=-Gene, names_to="Sample", values_to="Counts")

#Plot density distribution for all samples
ggplot(long_counts, aes(x=Counts, color=Sample)) +
 geom_density(alpha=0.5) +
 scale_x_log10() +
 theme_minimal() +
 labs(title="Global Count Distribution Across All Samples",
 x="Log10(Counts)",
 y="Density")+
 theme(legend.position="none")

#Save ggplot
output_dir <- "./"

ggsave(filename = file.path(output_dir, "Global_Count_Distribution.png"),
 width = 7,
 height = 5,
 dpi = 300)
```
```

★ Interpretation of Global Count Distribution Plot:

1. The density distribution plot for all samples follows a bimodal distribution (two peaks). Non-symmetric. Deviates from normal Gaussian bell curve.
2. This confirms non-normality.

4.5. SHAPIRO-TEST FOR ALL SAMPLES

What is the Shapiro-Wilk test?

The **Shapiro-Wilk test** is a statistical test used to determine if a dataset is normally distributed, by evaluating a single numeric variable against two hypotheses:

- **Null Hypothesis (H_0):** The data is normally distributed.
- **Alternative Hypothesis (H_a):** The data is not normally distributed.

Interpretation of the Shapiro-Wilk Test:

- If the $p > 0.05$ = Failure to reject null hypothesis. This means that the data is roughly normally distributed, and therefore, it is safe to use parametric tests like t-tests or ANOVA.
- $p \leq 0.05$ = Reject the null hypothesis. This means that the data significantly deviates from a normal distribution, and non-parametric alternatives like Wilcoxon or Mann-Whitney may be required.

4.5.1. What is normality?

This means that the data follows a normal (i.e., Gaussian) distribution, characterized by the symmetric “bell” curve, with the peak representing the mean of the data, where most of the data clusters. Outliers taper off equally in both directions.

Histograms can be used to visually inspect the normality of a dataset. If the shape is roughly bell-shaped, the data is normal.

The Shapiro-Wilk test is a statistical test used to determine if a dataset is normally distributed. If the $p > 0.05$ = Failure to reject null hypothesis. This means that the data is roughly normally distributed, and therefore, it is safe to use parametric tests like t-tests or ANOVA.

4.5.2. Normalization

A statistical method of bringing all of the data to a comparable scale, particularly in cases where the features on the dataset have a wide range, for example, a dataset looking at daily cell count in an in vitro organoid culture (ranging from 1,000 to 100,000 cells).

This is an issue for downstream analysis tools like KNN that rely on distance calculation.

```
```{r}
#Test if read counts are normally distributed
test_normality <- function(column_data) {
 set.seed(42)

 sample_size <- min(4000, length(column_data))
 sample_data <- sample(column_data, sample_size)
}
```

```

res <- shapiro.test(sample_data)
return(res$p.value)
}

#Apply function to every column (i.e., all samples)
all_p_values <- apply(count_matrix, MARGIN=2, FUN=test_normality)

#Convert to dataframe
normality_results <- data.frame(
 Sample=names(all_p_values),
 p_value=all_p_values,
 Is_Normal=ifelse(all_p_values>0.05, "Yes", "No (Reject Null Hypothesis)")
)

print(normality_results)
` ``

```

#### 4.5.3. Normality Test Results

Sample	p_value	Is_Normal	Accept/Reject
A778_D0	2.052916e-85	No	Reject Null Hypothesis
A778_D6	6.628042e-85	No	Reject Null Hypothesis
A820_D0	3.021601e-87	No	Reject Null Hypothesis
A820_D6	4.599661e-87	No	Reject Null Hypothesis
A870_D0	3.942424e-86	No	Reject Null Hypothesis
A870_D6	1.444317e-85	No	Reject Null Hypothesis
A899_D0	5.709202e-86	No	Reject Null Hypothesis
A899_D6	2.983315e-85	No	Reject Null Hypothesis

**Table 3. Normality testing of transcriptome read counts across HGSOC organoid samples.** Statistical evaluation of count distributions for four patient-derived lines (A778, A820, A870, and A899) at both immediate post-isolation baseline (Day 0) and following *in vitro* culture (Day 6). Extremely low p-values ( $p < 0.05$ ) across all samples dictate the rejection of the null hypothesis that the data follow a normal distribution. These findings confirm the non-Gaussian, highly skewed nature of raw RNA-Seq read counts, statistically justifying the downstream implementation of generalized linear models based on the Negative Binomial distribution within the DESeq2 framework rather than standard parametric methods.

#### ★ Interpretation of Normality Test Results:

1. All p-values are below 0.05 = Highly non-normal; rules out standard parametric tests.
2. Strongly reject the null hypothesis (that states that the data is normally distributed). This means the data deviates drastically from a Gaussian bell curve.
3. Rules out standard parametric tests such as t-test, ANOVA & standard linear regression, that assume symmetric distribution. Parametric tests

applied directly to raw count data would cause false positives to skyrocket, and statistical power to plummet.

4. Perfect for DESeq2 that models the data using a Negative Binomial distribution

#### 4.5.4. What is the p-value, generally?

As it relates to RNA-seq data and statistical tests, the p-value is a probability between 0 and 1 that measures the strength of the experimental results against the null hypothesis (i.e., baseline assumption).

If:

- $p > 0.05$  = Failure to reject null hypothesis (e.g., the gene's expression change is NOT statistically significant, and any variation could be due to "background noise").
- $p \leq 0.05$  = Reject the null hypothesis (e.g., the gene's expression change is statistically significant).

#### 4.6. DIFFERENTIAL GENE EXPRESSION ANALYSIS (DESeq2)(Paired Experimental Modeling)

##### 4.6.1. What is Differential Gene Expression Analysis?

DGEA is used to determine which genes show statistically significant changes in expression levels between different experimental conditions (for example, comparing the HGSOc organoids at Day 0 versus Day 6). DESeq2 normalizes the raw read counts using the **median-of-ratios method**. This accounts for differences in sequencing depth (library size) and library composition (where a few highly expressed genes can "soak up" all the reads, making other genes look falsely downregulated). RNA-Seq count data is highly skewed and cannot be modeled using a standard normal (bell-curve) distribution. Instead, DESeq2 models the counts using a **Negative Binomial distribution**.

```
``{r deseq2-setup, message=FALSE, warning=FALSE}

#Create metadata table using colnames
metadata <- data.frame(
 Sample_ID = colnames(count_matrix),
 Patient = factor(str_extract(colnames(count_matrix), "A[0-9]+")),
 Day = factor(if_else(str_detect(colnames(count_matrix), "D0"), "Day_0", "Day_6"),
 levels=c("Day_0", "Day_6"))
)

#Assign values from Sample_ID column as row names of metadata dataframe.
rownames(metadata) <- metadata$Sample_ID

#Verify that matrix columns align with metadata rows
stopifnot(all(colnames(count_matrix) == rownames(metadata)))

#Build dds design formula
```

```

dds <- DESeqDataSetFromMatrix(
 countData = count_matrix,
 colData = metadata,
 design = ~ Patient + Day
)

#Pre-filter for low-count genes (only keep genes with >= 10 reads in at least 4
samples)
keep <- rowSums(counts(dds) >= 10) >= 4
dds <- dds[keep,]

#Genes remaining after low-read pre-filtering
cat("Genes remaining after low-read pre-filtering:", nrow(dds), "out of",
nrow(count_matrix), "\n")

#Run DESeq (estimating size factors, estimating dispersions, gene-wise dispersion estimates, mean-dispersion
relationship, final dispersion estimates, fitting model and testing)
dds <- DESeq(dds)

'''

```

#### 4.7. PRINCIPAL COMPONENT ANALYSIS (PCA)

Principal Component Analysis (PCA) is a statistical technique used to reduce the number of features (i.e., dimensionality reduction) and visualize high-dimensional data in 2D or 3D. It does so by transforming a high-dimensional dataset with many correlated variables into a smaller set of uncorrelated variables called principal components, while reducing noise, removing multicollinearity, and preserving as much of the original data's variability as possible.

- **First Principal Component (PC1) (primary axis)** captures the maximum possible variance (or most dominant pattern) in the data.
- **Second Principal Component (PC2):** Captures the next highest variance and is perpendicular to PC1, ensuring that the information it captures is completely independent and does not overlap with the first component.

```

```{r}
#Variance Stabilizing Transformation (`vst`) (needed to execute a Principal
Component Analysis)
#For RNA-seq, PCA should be performed on variance-stabilized or transformed data
because raw
# counts are dominated by highly expressed genes.

vsd <- vst(dds, blind=FALSE)

#Create PCA data
pca_data <- plotPCA(vsd, intgroup = c("Patient", "Day"), returnData=TRUE)
percentVar <- round(100 * attr(pca_data, "percentVar"))

#Create PCA plot
pca_plot <- ggplot(pca_data, aes(x=PC1, y=PC2, color=Patient, shape=Day)) +

```

```

geom_point(size=4, alpha=0.9) +
labs(
  title= "Principal Component Analysis (PCA) of HGSOC Organoids",
  subtitle= "Evaluating variance structure across Patient Lines and Timepoints",
  x=paste0("PC1: ", percentVar[1], "% variance explained"),
  y=paste0("PC2: ", percentVar[2], "% variance explained")
) +
theme_minimal(base_size = 13) +
theme(plot.title=element_text(face="bold"))

print(pca_plot)

#Save PCA ggplot
ggsave(filename = file.path(output_dir, "HGSOC_PCA_Plot.png"),
  plot = pca_plot,
  width = 7,
  height = 5,
  dpi = 300)

...

```

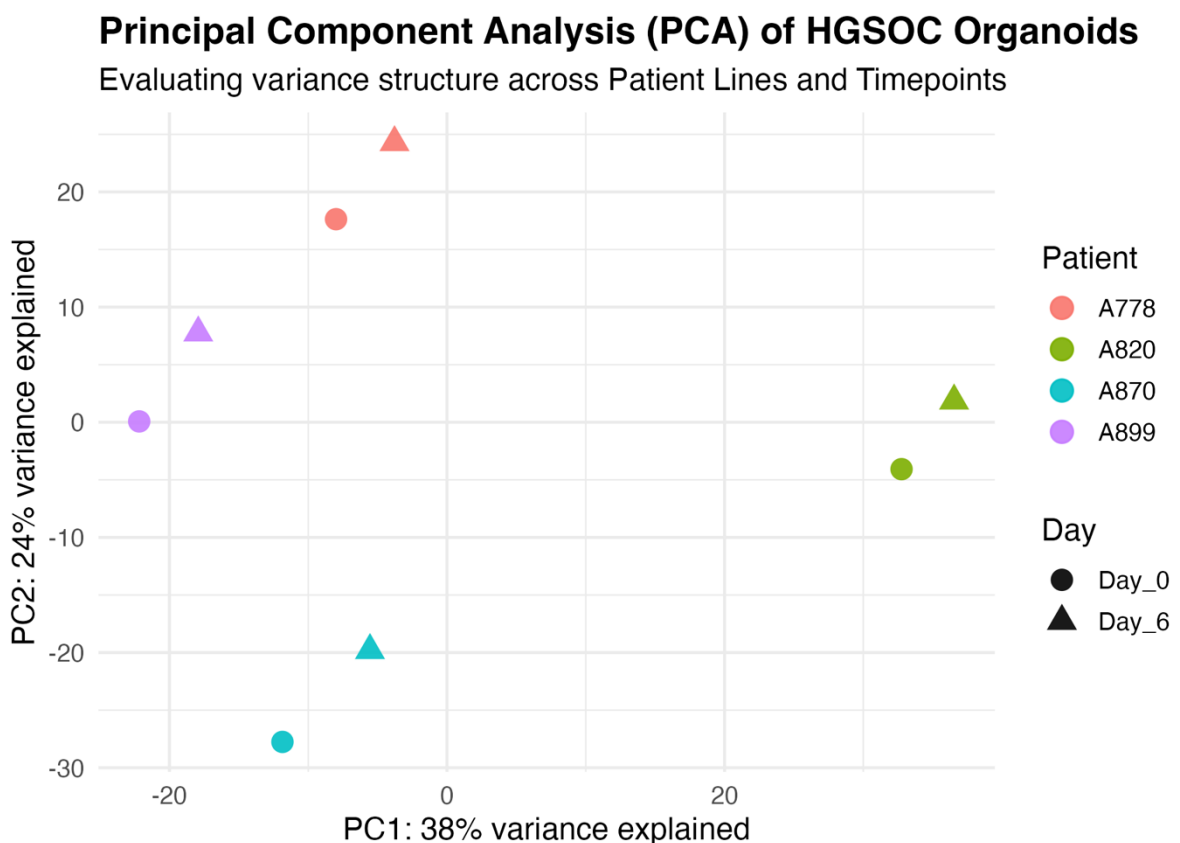


Figure 5. Principal Component Analysis (PCA) of HGSOC Organoids. PCA of variance-stabilized RNA-seq data showing global transcriptomic variation. Samples are coloured by patient and shaped by timepoint (Day 0 vs. Day 6). **Strong clustering by patient indicates that patient-specific differences are the primary source of variation.**

★ Interpretation of PCA Plot:

1. PC1 (38%) and PC2 (24%) explain 62% of the total variation in gene expression.
2. PC1 (38%) is likely driven by inter-patient genetic heterogeneity.
3. PC2 (24%) is likely driven by laboratory/in vitro adaptation and metabolic reprogramming.
4. Samples cluster primarily by patient, indicating patient-specific differences are the major source of variation.
5. Day 0 and Day 6 samples from the same patient remain closely grouped, showing good biological consistency.
6. All Day 6 samples shift upward along PC2, suggesting a shared temporal or treatment-related transcriptional response.
7. Gene expression changes occur between Day 0 and Day 6 in all patients, indicating a common transcriptomic response over time.
8. Patient-specific expression patterns are maintained over time, indicating organoids preserve the molecular characteristics of the original tumors.

4.8. EXTRACTION OF DIFFERENTIAL EXPRESSION RESULTS & VOLCANO PLOT

```
```{r}
#Extract Differential Expression results comparing Day6 v. Day0 baseline
res <- results(dds, contrast = c("Day", "Day_6", "Day_0"))

#Convert to tidy dataframe for ggplot compatibility
res_df <- as.data.frame(res) %>%
 mutate(Significance = case_when(
 padj < 0.05 & log2FoldChange > 1 ~ "Upregulated Genes (Day6)",
 padj < 0.05 & log2FoldChange < -1 ~ "Downregulated Genes (Day6)",
 TRUE ~ "Not Significant"
))

#Plot ggplot
volcano_plot <- ggplot(res_df, aes(x=log2FoldChange, y=-log10(padj),
color=Significance)) +
 geom_point(alpha = 0.6, size = 1.5) +
 scale_color_manual(values =c(
 "Upregulated Genes (Day6)" = "firebrick3",
 "Downregulated Genes (Day6)" = "dodgerblue3",
 "Not Significant" = "grey60"
)) +
 geom_hline(yintercept=-log10(0.05), linetype="dashed", color="black", alpha=0.7)
+
 geom_vline(xintercept= c(-1, 1), linetype="dashed", color="black", alpha=0.7) +
 labs(
 title= "Volcano Plot of Differential Gene Expression",
 subtitle= "HGSOC Patient Organoids: Day 6 v. Day 0 Baseline",
 x = "Log2 fold Change (LFC)",
 y = "-Log10 Adjusted p-value (FDR)"
) +
```



```

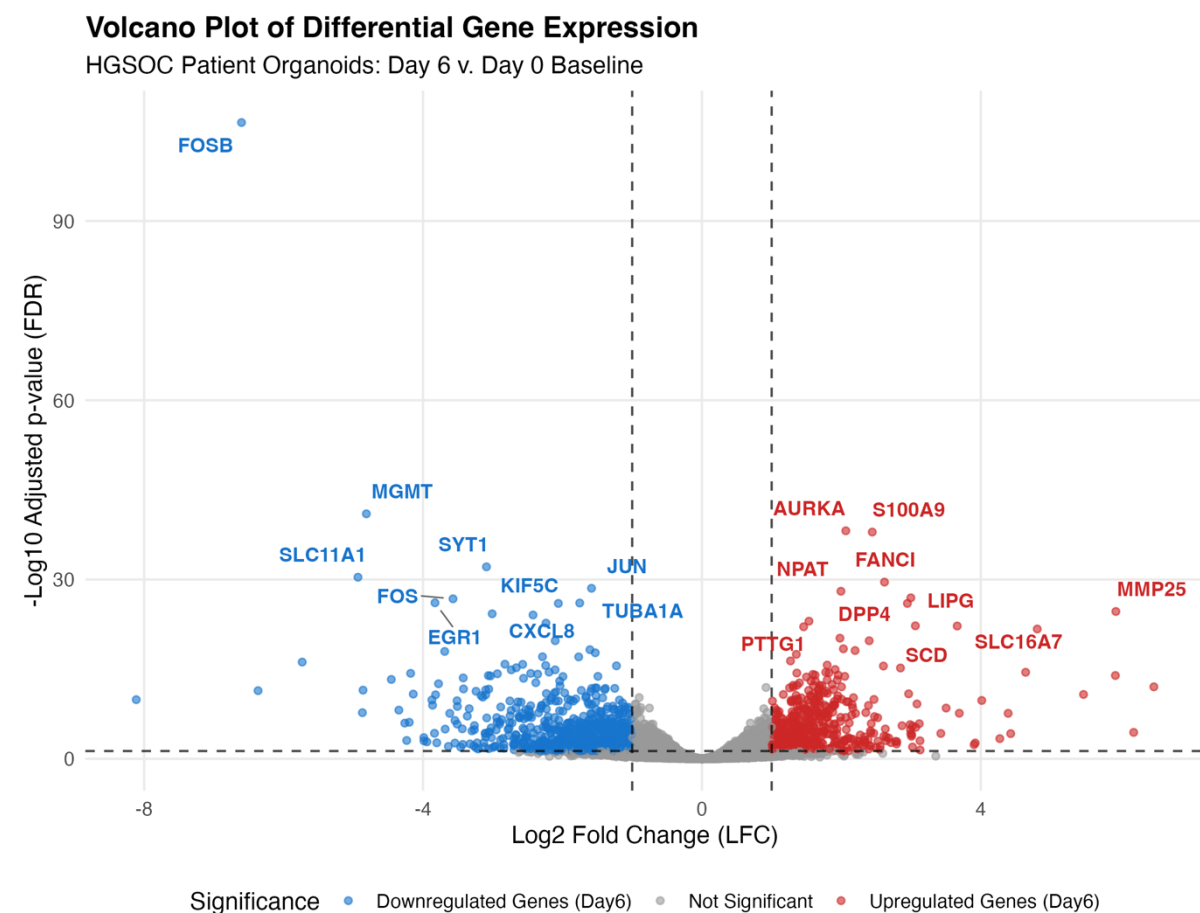
theme_minimal(base_size=13) +
theme(
 legend.position="bottom",
 plot.title = element_text(face="bold")
)

print(volcano_plot)

#Save ggplot
ggsave(filename = file.path(output_dir, "HGSOC_Volcano_Plot.png"),
 plot = volcano_plot,
 width = 7,
 height = 6,
 dpi = 300)

` ``

```



**Figure 6. Volcano Plot of Differential Gene Expression (Day 6 vs. Day 0 baseline) across all HGSOC patient-derived organoids.** Each point represents a gene, illustrating the overall transcriptomic changes associated with organoid culture over time.

### ★ Interpretation of Volcano Plot:

1. This plot is the result of a paired differential expression analysis across the entire cohort (i.e., all patient organoids (A778, A820, A870, and A899) analyzed simultaneously).
2. The plot groups all the Day 0 baseline samples together and compares them directly against all the Day 6 samples.
3. Shows the transcriptomic changes that happen to all HGSOC organoids as they transition from Day 0 to Day 6 in culture.
4. Each dot represents a gene's avg. expression across entire patient group.

### 4.9. TOP 100 DEGs HEATMAP

```
```{r}
#Subset the top 100 DEGs by significance
top_100_DEG <- res_df %>%
  filter(!is.na(padj)) %>%
  arrange(padj) %>%
  head(100) %>%
  rownames()

#Variance stabilization
vsd <- vst(dds, blind = FALSE)
heatmap_mat_100 <- assay(vsd)[top_100_DEG, ]

#Add column annotations
annotation_col <- data.frame(Day = dds$Day)
rownames(annotation_col) <- colnames(dds)

ann_colors <- list(
  Day = c(Day_0 = "grey80", Day_6 = "limegreen")
)

#Plot and Save Heatmap
pheatmap(
  mat = heatmap_mat_100,
  scale = "row",
  clustering_distance_rows = "euclidean",
  clustering_distance_cols = "euclidean",
  clustering_method = "complete",
  annotation_col = annotation_col,
  annotation_colors = ann_colors,
  show_colnames = FALSE,
  show_rownames = FALSE,
  legend = TRUE,
  annotation_legend = TRUE,
  main = "Top 100 Differentially Expressed Genes – Day 6 v. Day 0 Baseline",
  color = colorRampPalette(c("dodgerblue3", "white", "firebrick3"))(50),
  filename = file.path(output_dir, "HGSOC_Top_100_DEGs_Heatmap.png"),
  width = 6.5,
  height = 9.8
)
```

```
# custom_legend <- "Color Key Interpretation:\n• Deep Red (+2): High Expression\n(Upregulated / turned 'on' relative to the gene's average across all samples).\n• White (0): Average Expression (Baseline average level across all samples).\n• Deep Blue (-2): Low Expression (Downregulated / turned 'off' relative to the gene's\naverage)."\n# mtext(custom_legend, side = 1, line = -2, adj = 0, cex = 0.8, font = 3)\n# dev.off()\n```\n
```

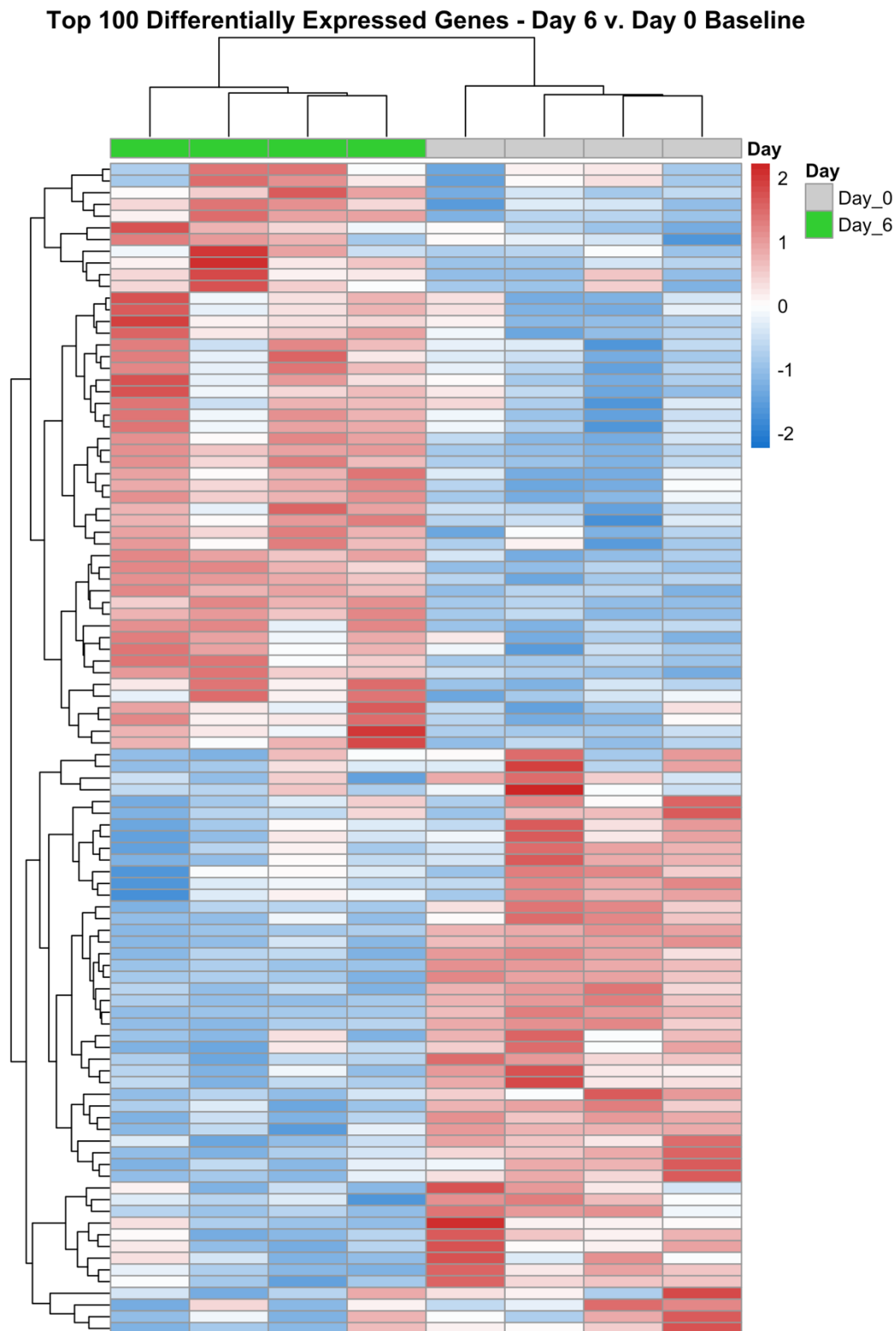


Figure 7. Heatmap of the Top 100 Differentially Expressed Genes (Day 6 vs. Day 0). Samples cluster perfectly by timepoint (top dendrogram). The genes split into two clear blocks: ~60% upregulated and ~40% downregulated at Day 6 (left dendrogram). Colours indicate relative expression variations (Z-scores). **Deep Red (+2)** = Upregulated, **White (0)** = Average Expression, **Deep Blue (-2)**: Downregulated.

★ Interpretation of Heatmap of the Top 100 Differentially Expressed Genes (Day 6 vs. Day 0):

1. Each column represents a patient's organoid sample.
2. Each row represents one of the top 100 most statistically significant genes determined DESeq2.
3. Dendrogram groups them based on transcriptomic similarity.
4. Left dendrogram = splits the 100 genes into two clear patterns, a top block of genes upregulated at Day 6 (~60%) and a bottom block of genes downregulated at Day 6 (~40%).
5. Top dendrogram = cluster samples perfectly by timepoint (Day 6 & Day 0)

★ Interpretation of Colour Key:

1. Colours represent relative expression variations (Z-scores).
2. Deep Red (+2): High Expression (Upregulated / turned 'on' relative to the gene's average across all samples).
3. White (0): Average Expression (Baseline average level across all samples).
4. Deep Blue (-2): Low Expression (Downregulated / turned 'off' relative to the gene's average).

4.9.1. What is the Z-score (standard score)?

The Z-score (standard score) describes a given data point's relationship to the mean of a group of values. It tells you how many standard deviations a single data point lies above or below the mean. The Z-score is a crucial tool for normalizing data, as it allows for comparison of scores from entirely different datasets or distributions on a level playing field.

Early-stage culture induced changes have been successfully characterized:

- **DGE results (Figures 6 and 7) characterize Transcriptomic Shift (Days 0–6).** Transition from stress/hypoxia survival (downregulated *FOS/FOSB/JUN/EGR1/CXCL8/MGMT/SLC11A1*) to active proliferation (upregulated *AURKA/PTTG1/FANCI/NPAT*).
- **PCA (Figure 5) characterizes how the culture shift interacts with the patient's original tumor biology, with patient-specific genetics remaining the dominant driver over time (PC1: 38% variance).**

4.10. SIGNIFICANCE EXTRACTION & MULTIPLE TESTING CORRECTION

```
```{r}
#Top 10 Upregulated Genes (i.e., those with the highest significance + positive LFC)
top_upreg_genes <- res %>%
 as.data.frame() %>%
 filter(padj < 0.05 & log2FoldChange > 1) %>%
 arrange(padj) %>%
 head(10)

#Top 10 Downregulated Genes (i.e., those with the highest significance + negative LFC)
```

```

top_downreg_genes <- res %>%
 as.data.frame() %>%
 filter(padj < 0.05 & log2FoldChange < -1) %>%
 arrange(padj) %>%
 head(10)

print(top_upreg_genes)
print(top_downreg_genes)

```

4.10.1. *TOP UPREGULATED GENES (at Day 6). These are genes that are turned "on" or significantly increased at Day 6 compared to Day 0.*

*Sorted from most significant (smallest  $p_{adj}$ ) to least significant.*

Gene	baseMean	log2FoldChange	lfcSE	stat	pvalue	Padj
ENSG00000087586.17 (AURKA)	1275.2146	2.063672	0.1508298	13.68213	1.298415e-42	6.986337e-39
ENSG00000163220.10 (S100A9)	2419.6719	2.443184	0.1793069	13.62571	2.816527e-42	1.136610e-38
ENSG00000134824.13 (FANCI)	3183.0934	2.618608	0.2165631	12.09166	1.168955e-33	2.695609e-30
ENSG00000170485.16 (NPAT)	2578.4804	1.993496	0.1692591	11.77778	5.081476e-32	9.113909e-29
ENSG00000101670.11 (LIPG)	393.2018	2.996363	0.2593585	11.55298	7.130573e-31	1.151017e-27
ENSG00000197635.9 (DPP4)	222.3187	2.946951	0.2600788	11.33099	9.215300e-30	9.916891e-27
ENSG00000100985.7 (MMP25)	378.9614	5.936808	0.5372668	11.05002	2.191719e-28	2.211170e-25
ENSG00000164611.12 (PTTG1)	912.2410	1.535121	0.1435565	10.69350	1.091689e-26	9.274756e-24
ENSG00000099194.5 (SCD)	8434.6822	3.061301	0.2910823	10.51696	7.216472e-26	5.547061e-23
ENSG00000112494.9 (SLC16A7)	133.2646	3.661387	0.3485185	10.50557	8.142575e-26	5.974429e-23

**Table 4. Top differentially expressed genes in HGSOC 3D organoids comparing Day 6 to Day 0 cultures.** A subset of the most highly significant differentially expressed genes sorted by statistical relevance. For each Ensembl gene ID and corresponding gene symbol, metrics computed via DESeq2 are provided: **baseMean** (mean of normalized counts across all samples);

**log2FoldChange** (magnitude and direction of expression change at Day 6 relative to Day 0); **lfcSE** (standard error estimate for the log<sub>2</sub> fold change); **stat** (Wald statistic value); **pvalue** (unadjusted Wald test p-value); and **p<sub>adj</sub>** (Benjamini-Hochberg adjusted p-value controlling for false discovery rate). Positive log<sub>2</sub> fold change values indicate robust transcriptional upregulation following the 6-day *in vitro* culture period.

4.10.2. *TOP DOWNREGULATED GENES (at Day 6). These are genes that are turned "off" or significantly decreased at Day 6 compared to Day 0.*

*Sorted from most significant (smallest p<sub>adj</sub>) to least significant.*

Gene	baseMean	log2FoldChange	lfcSE	stat	pvalue	p <sub>adj</sub>
ENSG00000125740.13 (FOSB)	1729.7785	-6.603510	0.2944237	-22.42859	2.070793e-111	3.342673e-107
ENSG00000085741.12 (MGMT)	351.9082	-4.812620	0.3394333	-14.17840	1.246603e-45	1.006133e-41
ENSG00000149043.16 (SYT1)	676.5197	-3.090101	0.2453940	-12.59241	2.324779e-36	7.505316e-33
ENSG00000018280.16 (SLC11A1)	130.4029	-4.933332	0.4024633	-12.25784	1.525018e-34	4.102808e-31
ENSG00000171223.5 (JUN)	2533.9503	-1.583497	0.1332506	-11.88360	1.440292e-32	2.906150e-29
ENSG00000170345.9 (FOS)	2159.0696	-3.570071	0.3101179	-11.51198	1.148079e-30	1.684753e-27
ENSG00000120738.7 (EGR1)	1513.4489	-3.828581	0.3366688	-11.37195	5.768308e-30	7.759336e-27
ENSG00000167552.13 (TUBA1A)	2211.9396	-1.752896	0.1543408	-11.35730	6.821759e-30	8.470526e-27
ENSG00000124249.6 (KIF5C)	594.7514	-2.059304	0.1816870	-11.33435	8.868938e-30	9.916891e-27
ENSG00000203747.10 (CXCL8)	185.5941	-3.008492	0.2744061	-10.96365	5.714921e-28	5.426486e-25

**Table 5. Top significantly downregulated genes in HGSOC 3D organoids following Day 6 culture.** A subset of the most highly significant differentially expressed genes sorted by statistical relevance. For each Ensembl gene ID and corresponding gene symbol, metrics computed via DESeq2 are provided: **baseMean** (mean of normalized counts across all samples); **log2FoldChange** (magnitude and direction of expression change at Day 6 relative to Day 0); **lfcSE** (standard error estimate for the log<sub>2</sub> fold change); **stat** (Wald statistic value); **pvalue** (unadjusted Wald test p-value); and **p<sub>adj</sub>** (Benjamini-Hochberg adjusted p-value controlling for false discovery rate). Negative log<sub>2</sub> fold change values indicate robust transcriptional downregulation or suppression following the 6-day *in vitro* culture period relative to the immediate post-isolation baseline (Day 0).

#### 4.11. Top highly significant genes and their associated cellular processes.

##### 4.11.1. Top Upregulated Genes at Day 6

At Day 6, the top upregulated genes were *AURKA*, *PTTG1*, *FANCI* and *NPAT*. As expected, these genes are associated with cell proliferation, DNA replication, biomass production and cell cycle progression, consistent with their adaptation to the non-tumor, stress-free, *in vitro* microenvironment.

Gene	Description/Associated Cellular Process(es)	Citations
<b><i>AURKA (Aurora Kinase A), PTTG1 (Securin)</i></b>	Critical master regulators of mitosis, spindle assembly, and chromosome segregation. Their upregulation indicates that by Day 6, the HGSOC organoids have entered a highly active proliferative phase in culture. Cell Cycle Progression; Cell proliferation.	(Blengini et al, 2025)(Guarino Almeida et al., 2020)(Vergani e al., 2025)
<b><i>FANCI</i></b>	DNA Replication via Fanconi anemia pathway; Cell Proliferation.	(Moriel-Carretero et al., 2017)
<b><i>NPAT</i></b>	Cell Cycle Progression; DNA Replication; Biomass Production.	(Gao et al., 2003)

**Table 6. Functional annotation and cellular processes of key upregulated genes in HGSOC 3D organoids.** A summary of select highly significant differentially expressed genes identified in the DESeq2 pipeline. For each gene symbol, its biological description, associated cellular processes, and supporting literature citations are detailed. The prominence of master regulators controlling mitosis, spindle assembly, DNA replication, and cell cycle progression (*AURKA*, *PTTG1*, *FANCI*, and *NPAT*) highlights a functional transition toward a highly active proliferative phenotype in the patient-derived HGSOC organoids by Day 6 of *in vitro* culture.

##### 4.11.2. Top Downregulated Genes at Day 6

At Day 6, the top downregulated genes were *FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8 (IL-8)*, *MGMT* and *SLC11A1*. As expected, these genes are associated with cell survival signaling, hypoxia responses, inflammation, or apoptosis, and immediate stress response, consistent with the *in vitro* tumor microenvironment, where HGSOC cells experience low oxygen (hypoxia), nutrient gradients, stromal fibroblasts, immune cells, endothelial cells, cytokines and growth factors.

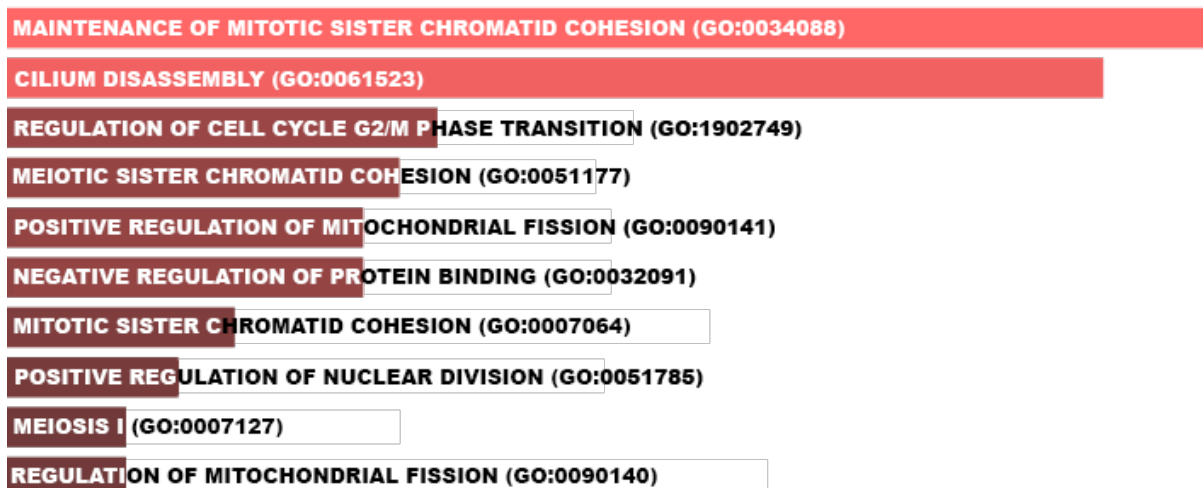


Gene	Description/Associated Cellular Process(es)	Citations
<b><i>FOS, FOSB, JUN, and EGR1</i></b>	Classic Immediate Early Genes (IEGs) and components of the AP-1 transcription factor complex. They spike dramatically whenever cells are physically disturbed, mechanically stressed, or newly isolated. Apoptosis modulation = <i>JUN, FOS, EGR1</i> ; hypoxia-related = <i>EGR1, FOS, JUN</i> ; survival signaling = <i>FOSB, FOS, MGMT, JUN, EGR1, CXCL8</i> .	(National Center for Biotechnology Information, 2026)(Ip and Davis, 1998)(Wisdom et al., 1999)(Zou and Zeng, 2023)(Rasras et al., 2024)(Wang et al., 2021)
<b><i>CXCL8 (IL-8)</i></b>	A major pro-inflammatory chemokine. Downregulation suggests inflammatory stress signaling has leveled off. Anti-apoptotic; hypoxia-related; survival signaling. Promotes proliferation, angiogenesis, migration, invasion and cell survival.	(Cambier et al., 2023) (Asokan and Bandapalli, 2021)
<b><i>MGMT</i></b>	Survival signalling; anti-apoptotic/pro-survival	(Fang, 2024)
<b><i>SLC11A1</i></b>	Iron metabolism; cellular homeostasis; oncogenesis; modulation of tumor environment; therapy resistance; tumor initiation; tumor progression; metastasis	(Banerjee et al., 2025)

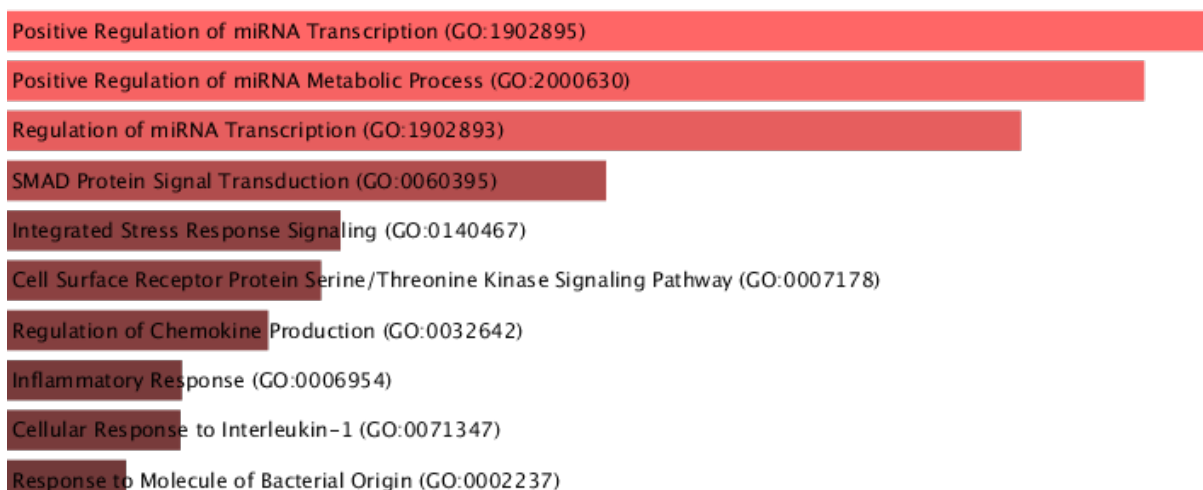
**Table 7. Functional annotation and cellular processes of key downregulated genes in HGSOc 3D organoids.** A summary of select highly significant differentially expressed genes showing negative fold changes in the DESeq2 pipeline. For each gene symbol, its biological description, associated cellular processes, and supporting literature citations are detailed. The marked downregulation of classic Immediate Early Genes (*FOS, FOSB, JUN, and EGR1*) alongside major pro-inflammatory markers like *CXCL8 (IL-8)* suggests that the acute cellular distress, mechanical isolation stress, and inflammatory signaling characteristic of the baseline, freshly isolated state (Day 0) have successfully leveled off or stabilized by Day 6 of *in vitro* 3D culture.

#### 4.12. Gene Ontology Enrichment Analysis (Day 6)

To identify enriched biological processes and pathways, Gene Ontology (GO) enrichment analysis was performed separately on the significantly upregulated ( $\log_2FC > 0.58$ ,  $p_{adj} < 0.05$ ) and downregulated ( $\log_2FC < -0.58$ ,  $p_{adj} < 0.05$ ) gene sets using the Enrichr web tool (<https://maayanlab.cloud/Enrichr/>). The GO Biological Process was utilized to identify biological process associated with each gene set. Statistical significance was adjusted for multiple testing using the Benjamini-Hochberg procedure, with enriched pathways defined at an adjusted threshold of  $q < 0.05$ .



**Figure 10. Gene Ontology Enrichment Analysis of Upregulated Genes (Day 6).** Top upregulated biological processes, showing exit from G0 (cilium disassembly) and progression into active mitosis (chromosome cohesion, G2/M transition). Essentially, The cells exited quiescence and entered a phase of rapid proliferation.



**Figure 11. Gene Ontology Enrichment Analysis of Downregulated Genes (Day 6).** Enriched GO terms confirm the loss of host microenvironmental stress and immune signaling, characterized by significant reductions in integrated stress response signaling, inflammatory response, and chemokine production. This confirms the exit of the cells from an acute inflammatory and stressed state (characteristic of patient malignant effusion fluid).

#### 4.13. UNSUPERVISED CLUSTERING & QUALITY CONTROL

Variance Stabilization Transformation/Testing (VST) is a normalization transformation method used to adjust gene expression data so that very highly expressed genes do not overpower the analysis. This gives all genes a more equal chance to contribute, making patterns between samples easier to identify. Simply, VST ensures that the variance is not dominated by high-expression genes, and that

the data is homoscedastic (i.e., has a constant variance across the entire range of mean values). This prevents highly expressed genes from dominating downstream analyses (in this case the quality control PCA below).

```
```\r}
# Apply Variance Stabilization Transformation/Testing (VST) (i.e., normalize the
data so that the variance is not dominated by high-expression genes). This is
essential for plotting QC PCA below.
vsd <- vst(dds, blind = FALSE)

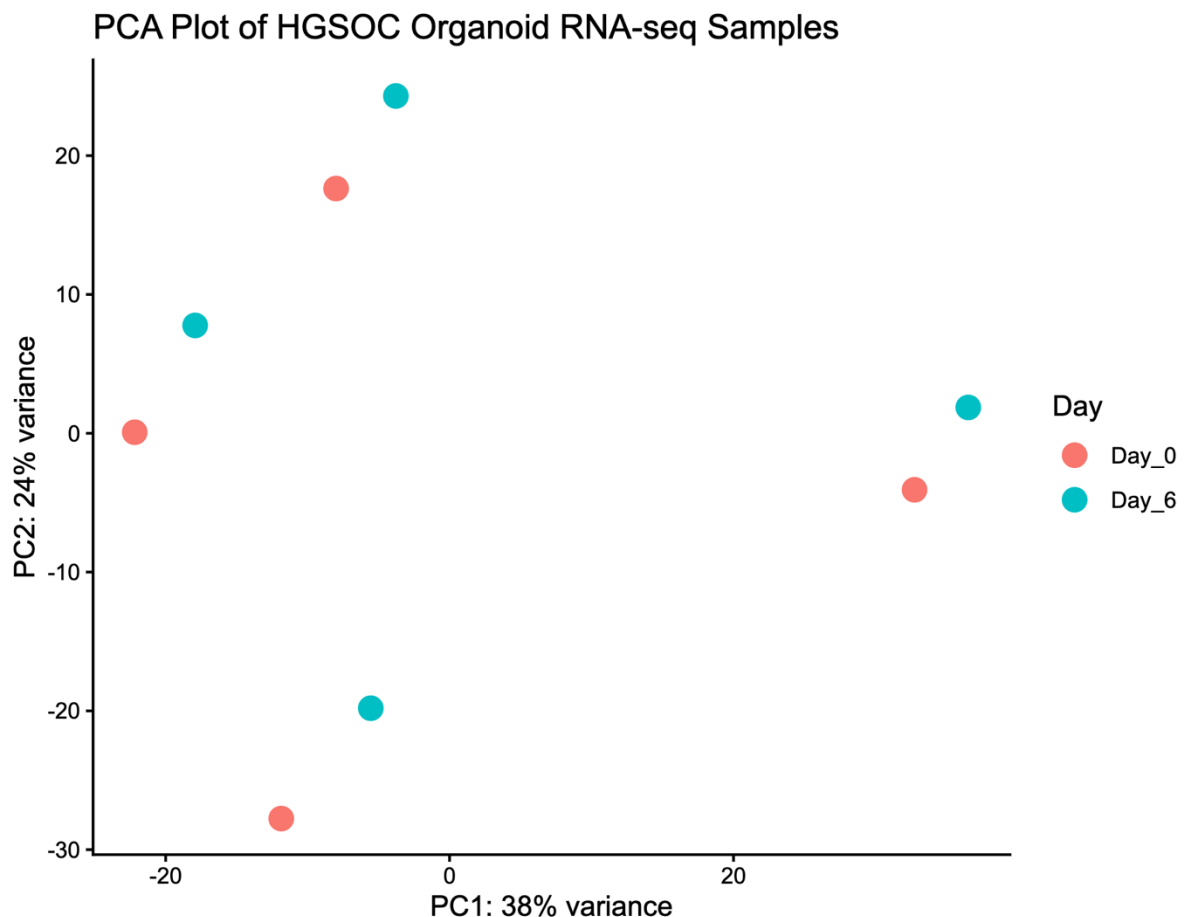
vsd_mat <- assay(vsd) #transformed expression matrix

# Plot PCA to assess global transcriptomic variation
pcaData <- plotPCA(
  vsd,
  intgroup = "Day",
  returnData = TRUE
)

#Calculate PCA1 and PC2 percentage variance
percentVar <- round(100 * attr(pcaData, "percentVar"))

#Plot QC PCA
qc_pca <- ggplot(pcaData, aes(
  x = PC1,
  y = PC2,
  color = Day
)) +
  geom_point(size = 4) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  ggtitle("PCA Plot of HGSOC Organoid RNA-seq Samples") +
  theme_classic()

print(qc_pca)
```\r
```



**Figure 8. QC Principal Component Analysis (PCA) of HGSOC organoid transcriptomes.** Variance-stabilized RNA-Seq read counts across the four patient-derived lines are projected onto the first two principal components, which together account for 62% of the total transcriptional variance (PC1: 38%; PC2: 24%). Individual samples are colour-coded by culture timepoint: immediate post-isolation baseline (Day 0, salmon) and after a 6-day *in vitro* culture period (Day 6, cyan). In alignment with the hierarchical clustering results, samples group tightly into four distinct coordinate spaces corresponding to individual patient genetic backgrounds. The proximity of Day 0 and Day 6 pairs for each unique patient line relative to the vast spatial separation between patient groups demonstrates that inter-patient biological heterogeneity remains the primary driver of transcriptomic variation, exceeding the gene expression shifts induced by the 6-day culture period.

4.13.1. How does this QC PCA differ from the DGE PCA plot?

## Summary Comparison

Feature	Figure 5 (DGE PCA Plot)	Figure 8 (QC PCA Plot)
Primary Visual Grouping	Coloured by <b>Patient ID</b>	Coloured by <b>Culture Day</b>
Core Question	"Are patient lines transcriptionally distinct?"	"Does culture timepoint override patient identity?"

<b>Key Takeaway</b>	Yes; samples cluster tightly by patient line.	No; Day 0 and Day 6 pairs stay together, showing high culture fidelity.
---------------------	-----------------------------------------------	-------------------------------------------------------------------------

In RNA-seq workflows, a good quality control PCA plot for this type of experiment aims to confirm two key things:

1. **High Culture Fidelity.** QC PCA Plot confirms that 6 days in culture does not dramatically alter the baseline global transcriptome or introduce a massive technical batch effect.
2. **Preservation of Biological Identity.** QC PCA Plot proves that sample identity (patient line) remains the dominant biological signal throughout the experiment.

The QC PCA Plot highlights global stability across conditions, showing that patient-specific biology far outweighs any transcriptional shift caused by the 6-day culture period.

```

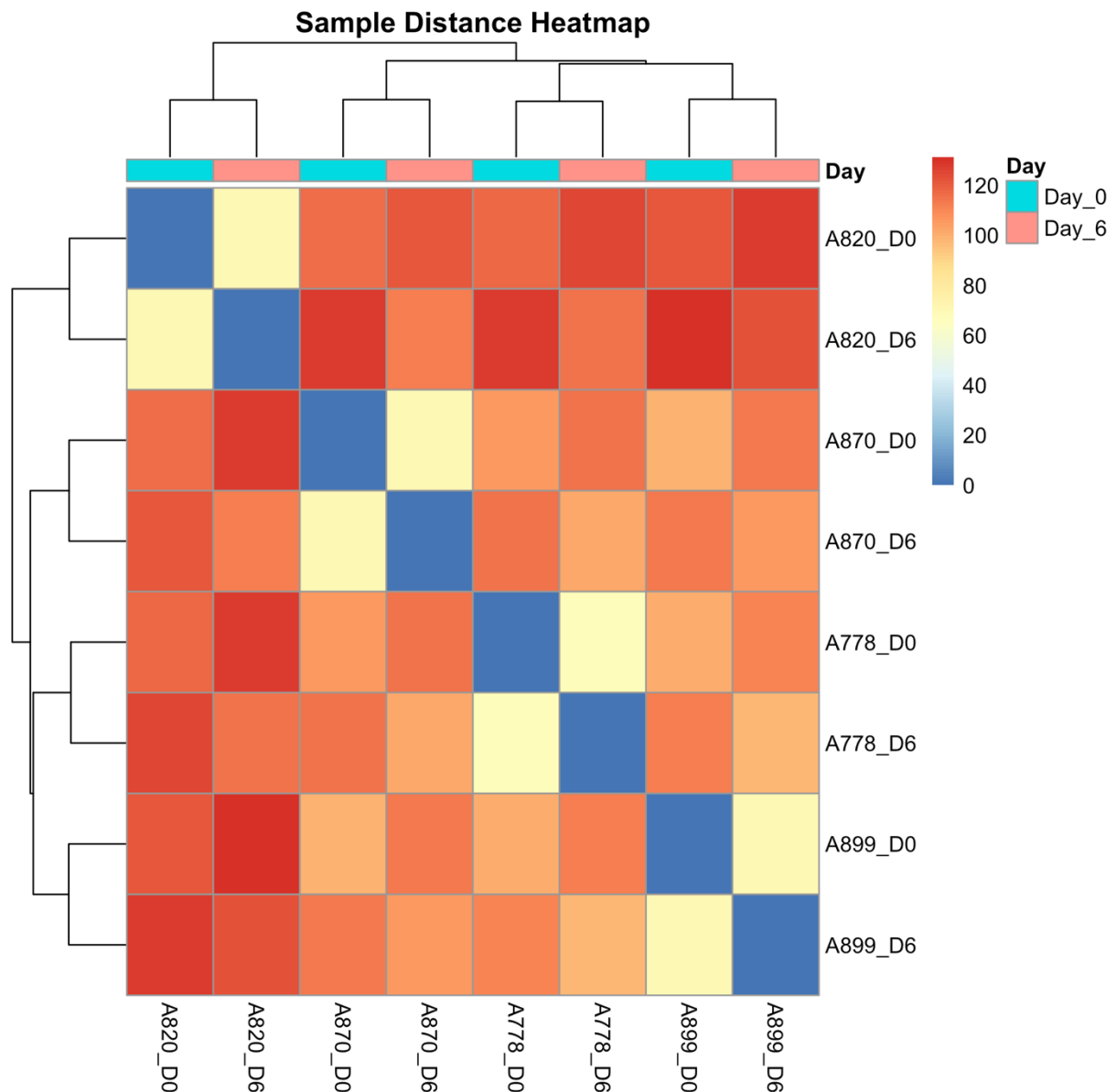
```{r}
# Create Sample Distance Matrix (for following heatmap)
sampleDists <- dist(t(assay(vsd))) #calculating math distance between entire
transcriptomes will cause mislabeled samples to stick out.
sampleDistMatrix <- as.matrix(sampleDists)

rownames(sampleDistMatrix) <- colnames(vsd)
colnames(sampleDistMatrix) <- colnames(vsd)

# Heatmap visualizing sample similarity.
# As opposed to Top 100 DEGs heatmap (supervised), this sample distance heatmap
(unsupervised) measures all genes in this cohort.

pheatmap(
  sampleDistMatrix,
  annotation_col = as.data.frame(colData(vsd)[, "Day", drop = FALSE]),
  main = "Sample Distance Heatmap",
  filename = file.path(output.dir, "Sample_Distance_Heatmap.png")
)
```

```



**Figure 9. Sample distance heatmap of HGSOc organoid transcriptomes across experimental conditions.** Hierarchical clustering analysis based on Euclidean distances computed from variance-stabilized RNA-Seq read counts across four patient-derived lines (A778, A820, A870, and A899). The colour gradient indicates relative distance, where darker blue signifies high transcriptional similarity (lower distance) and red signifies lower similarity (higher distance). The dendrogram demonstrates that samples cluster primarily by patient genetic background rather than by treatment timepoint (Day 0, cyan; Day 6, salmon). For each independent patient line, the Day 0 and Day 6 cultures cluster directly together as immediate sister pairs, highlighting that inter-patient biological variability outweighs the transcriptomic shifts induced by the 6-day *in vitro* culture microenvironment.

★ **Interpretation of QC Sample Distance Heatmap:**

1. Colours represent the Euclidean distance (i.e., mathematical distance between transcriptomes)

2. Demonstrates sample similarity, clustering of replicates, and absence of obvious outliers.
3. Individual patient organoids cluster strictly with their own matching timepoint (e.g., A820\_D0 pairs with A820\_D6). The tight pairing proves that the organoids successfully retain their unique, patient-specific molecular characteristics over time in the lab.
4. This heatmap is perfect for QC because, the perfect grouping of patient pairs proves excellent data integrity with zero sample tracking mix-ups or cross-contamination during lab or sequencing steps.
5. Experimental groups maintain a clean, global transcriptomic separation.
6. The solid blue diagonal line passing through the matrix represents a mathematical distance of 0, where each sample is compared directly to itself.
7. The yellow squares immediately off the diagonal line show that a patient's Day 0 and Day 6 samples remain highly similar to one another.
8. The large orange and dark red blocks signify high mathematical distance (100–120), highlighting just how distinct the different patient tumor lines are from each other.

4.13.2. *How does this sample distance heatmap differ from the DGE Top 100 heatmap?*

| Feature / Metric                | DGE Top 100 Heatmap<br>(Figure 7)                                                                                                                                                                                                                       | Sample Distance Heatmap<br>(Figure 9)                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analysis Type</b>            | Supervised (biased; genes pre-selected based on statistical significance)                                                                                                                                                                               | Unsupervised (unbiased; uses no experimental group assumptions)                                                                                                                                                                                                                 |
| <b>What the Data Represents</b> | Focuses specifically on the top 100 genes that change most significantly between Day 0 and Day 6. By stripping away all the genes that remain unchanged across patient backgrounds, it isolates the specific response to the 6-day culture environment. | Measures overall transcriptomic distance across all genes simultaneously. It compares how similar Sample A's entire profile is to Sample B's. Because it evaluates the whole transcriptome, the baseline expression driven by patient genetics completely dominates the signal. |
| <b>Primary Clustering</b>       | Clusters primarily by experimental timepoint (Day 0 vs. Day 6)                                                                                                                                                                                          | Clusters primarily by patient genetic background (baseline genetics)                                                                                                                                                                                                            |
| <b>Clustering Pattern Flip</b>  | By filtering for genes selected specifically because they differ between timepoints, the "patient background noise" is removed. As a result, all Day 6 samples cluster together on the left, and all Day 0 samples cluster together on the right.       | Inter-patient variability outweighs the culture effect. Patient lines (e.g., A820 Day 0 and Day 6) remain closer to each other than to any other patient's cells at the same timepoint.                                                                                         |

|                                   |                                                                                                                                                         |                                                                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>What the Colours Represent</b> | Relative Gene Expression (Z-score from -2 to +2)<br>• Deep red: High/upregulated expression<br>• Deep blue: Low/downregulated expression                | Euclidean Distance (0–120)<br>• Deep blue: High similarity (low distance)<br>• Orange/red: Low similarity (high distance)  |
| <b>Primary Purpose</b>            | <b>Biological Interpretation.</b><br>Visualize coordinated gene expression changes associated with metabolic reprogramming and cellular transition      | <b>Quality Control (QC).</b> Verify sample identity, detect outliers, and assess patient-specific transcriptomic stability |
| <b>Key Biological Takeaway</b>    | Transition to culture relieves the in vivo stress program (downregulated genes) while activating proliferation-associated pathways (upregulated genes). | Despite adaptation to culture, patient genetic identity remains the dominant global transcriptomic signature over time.    |

**Table 9.** The Sample Distance Heatmap is an unbiased sample-vs-sample metric showing that patients cluster with themselves globally. The DGE Top 100 Heatmap is a biased gene-vs-sample matrix showing exactly how the timepoints separate based on the most significant adaptive markers.

★ **Combined Interpretation of both PCA plots:**

- While each patient line retains its overall unique genetic identity (Figure 9), all four lines share a consistent, conserved biological reaction to the 6-day culture condition across a core set of genes (Figure 7).

## 5. Biological Interpretation

Early in vitro cultivation of HGSOc organoids triggers a significant transcriptomic shift/metabolic reprogramming. This rapid adaptation alters gene expression profiles and activates pathways tied to cell proliferation, cellular survival, and tumor biology.

**At Day 6,** The **upregulated** genes (AURKA, PTTG1, FANCI, and NPAT) are known to drive cell cycle progression and DNA replication (Blengini et al., 2025)(Guarino Almeida et al., 2020)(Vergani et al., 2025)(Moriel-Carretero et al., 2017)(Gao et al., 2003).

**At Day 6,** The **downregulated** genes (*FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8*, *MGMT*, *SLC11A1*) are known to suppress hypoxia, inflammation, and stress pathways (National Center for Biotechnology Information, 2026)(Ip and Davis, 1998)(Wisdom et al., 1999)(Zou and Zeng, 2023)(Asokan and Bandapalli, 2021)(Cambier et al., 2023)(Rasras et al., 2024)(Wang et al., 2021)(Banerjee et al., 2025)(Fang, 2024).



## 6. Conclusion

The hypothesis is supported. Transferring HGSOC cells from hypoxic malignant effusions to nutrient-rich culture triggers a transcriptomic shift from stress response to active proliferation.

By day 6, the short-term transition of HGSOC cells from a nutrient-stressed, hypoxic, malignant effusion environment to an *in vitro* nutrient-rich environment triggered a significant transcriptomic shift, relieving hypoxia signalling, downregulating genes associated with survival signalling pathways and anti-apoptosis, and upregulating pathways associated with cell proliferation. This adaptation is driven by a process known as metabolic reprogramming, where cellular metabolism is redirected to support proliferation rather than stress adaptation, and confirms that the organoids successfully stabilized.

Statistical modeling via DESeq2 successfully characterized this transition, revealing a downregulation of genes associated with hypoxia responses, inflammation, apoptosis, and other immediate stress response (such as *FOS*, *FOSB*, *JUN*, *EGR1*, *CXCL8 (IL-8)*, *MGMT*, and *SLC11A1*), and an upregulation of genes associated with cell proliferation, DNA replication, biomass production and cell cycle progression (such as *AURKA (Aurora Kinase A)*, *PTTG1 (Securin)*, *FANCI* and *NPAT*).

Characterizing this transcriptomic shift is vital because it proves that the *in vitro* organoid models are adapting, as opposed to mutating. It confirms that while the organoids successfully stabilize *in vitro*, they still preserve the patient's unique tumor architecture and biology (i.e., genetic, transcriptomic, and molecular profile). HGSOC organoids remain reliable patient-specific models, with individual genetic background remaining the dominant driver of expression variance (PC1: 38%). This interpatient heterogeneity is also confirmed by the heatmap.

This study proves that HGSOC organoids undergo predictable adaptation rather than random genetic divergence, and validates their integrity as premier functional platforms for high-throughput drug screening and precision oncology.

## 7. Future Studies

Future studies could reconstitute the tumor microenvironment by co-culturing organoids with autologous immune cells and fibroblasts under controlled hypoxic conditions. Testing if this rescues baseline stress genes (such as *FOS*, *JUN* and *CXCL8*) will create an immunologically faithful platform for evaluating novel immunotherapies.

## 8. AI DECLARATION

**ChatGPT was utilized to:**

1. Troubleshoot installing packages.
2. Convert Ensembl gene IDs (e.g., ENSG00000087586.17) to gene symbols (e.g., *AURKA*).
3. Refine code comments.

4. Get a better understanding of R functions such as "colorRampPalette", "set.seed", and "purrr::reduce".

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## 10. Author Declaration

Author has proofread, edited and agrees to the published version of this manuscript and accompanying poster.

## 11. Funding

This research received no external funding.

**Institutional Review Board Statement:** Not applicable. This study did not involve humans or animals.

**Informed Consent Statement:** Not applicable. This study did not involve humans or animals.

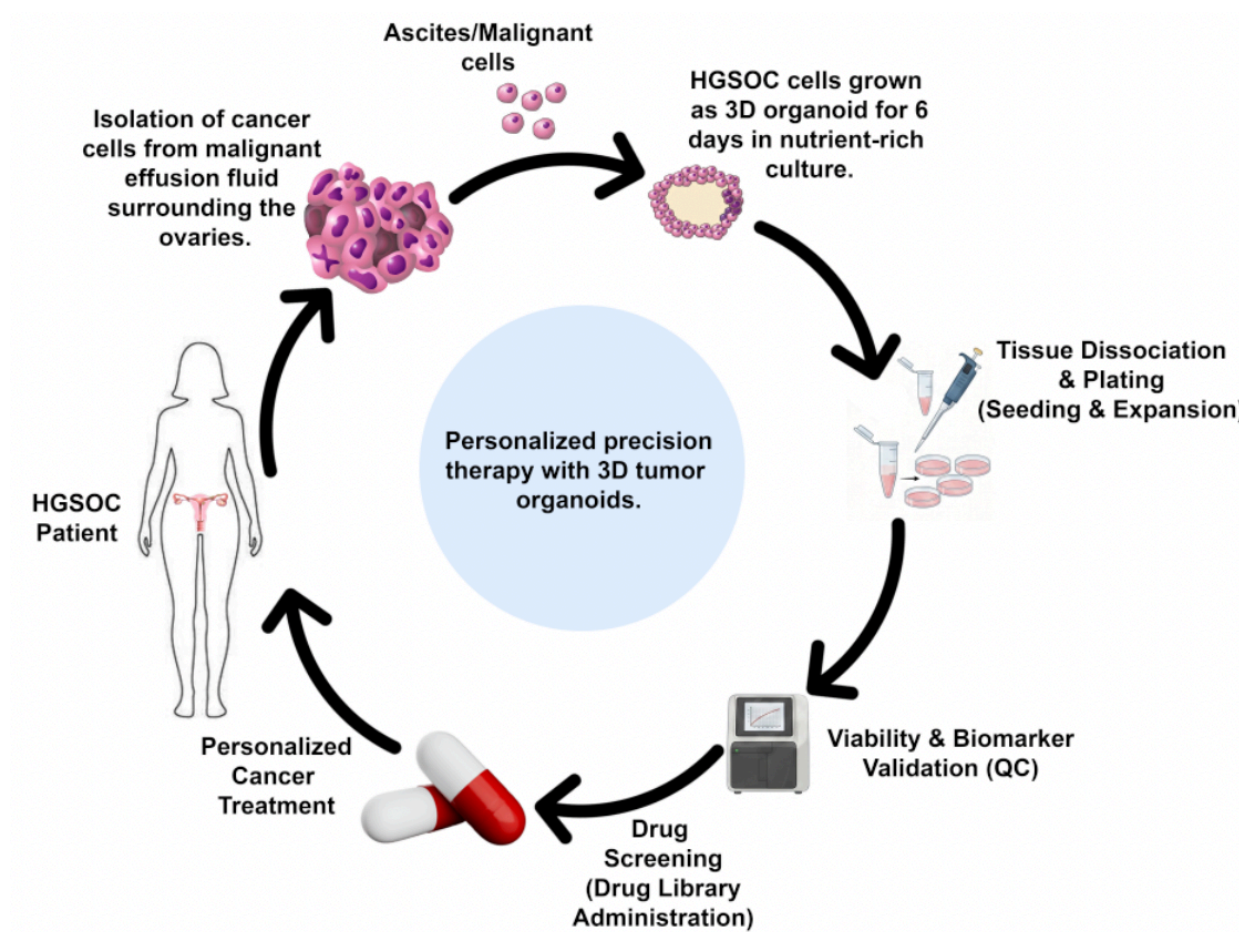
**Data Availability Statement:** Data sharing is not applicable to this article. No new data were created or analyzed in this study.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## 12. Acknowledgements

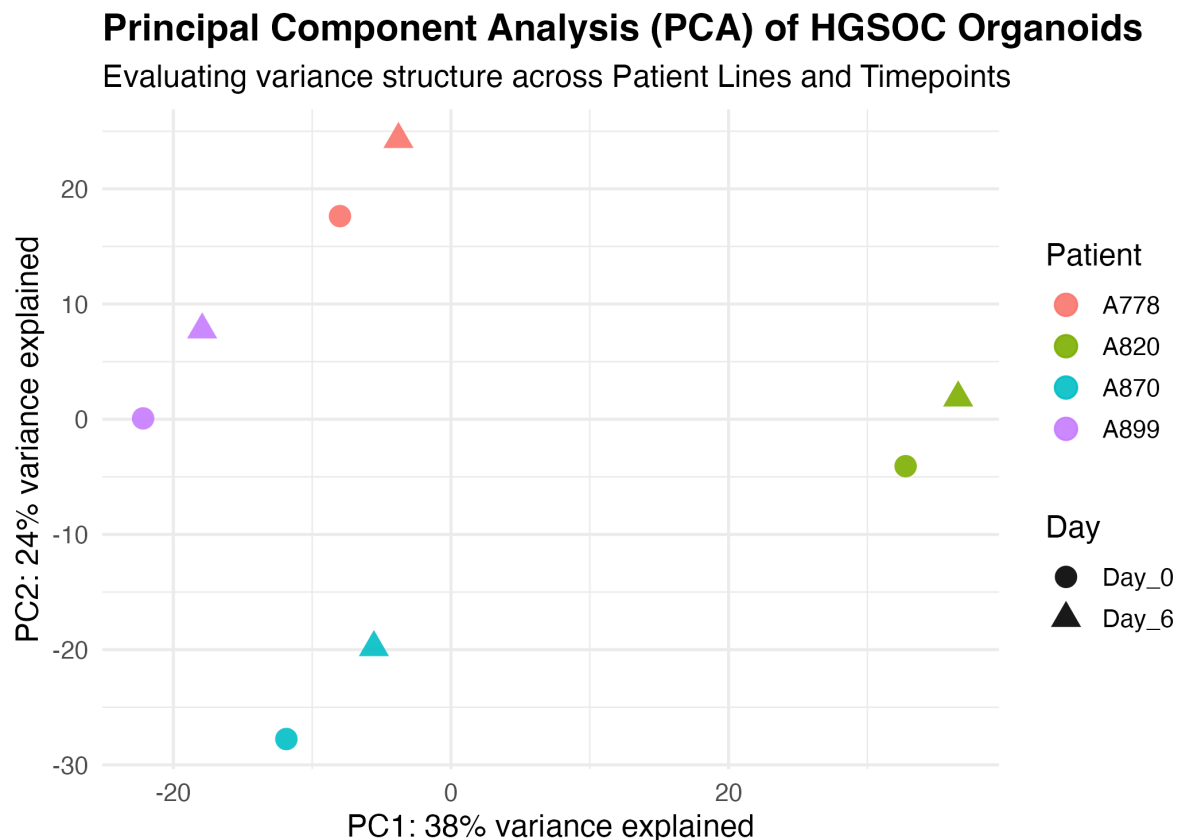
We would like to thank Professor Bolanos and our classmates in BINF5503 for their invaluable guidance, support, and constructive feedback throughout this project.

## 13. Appendix

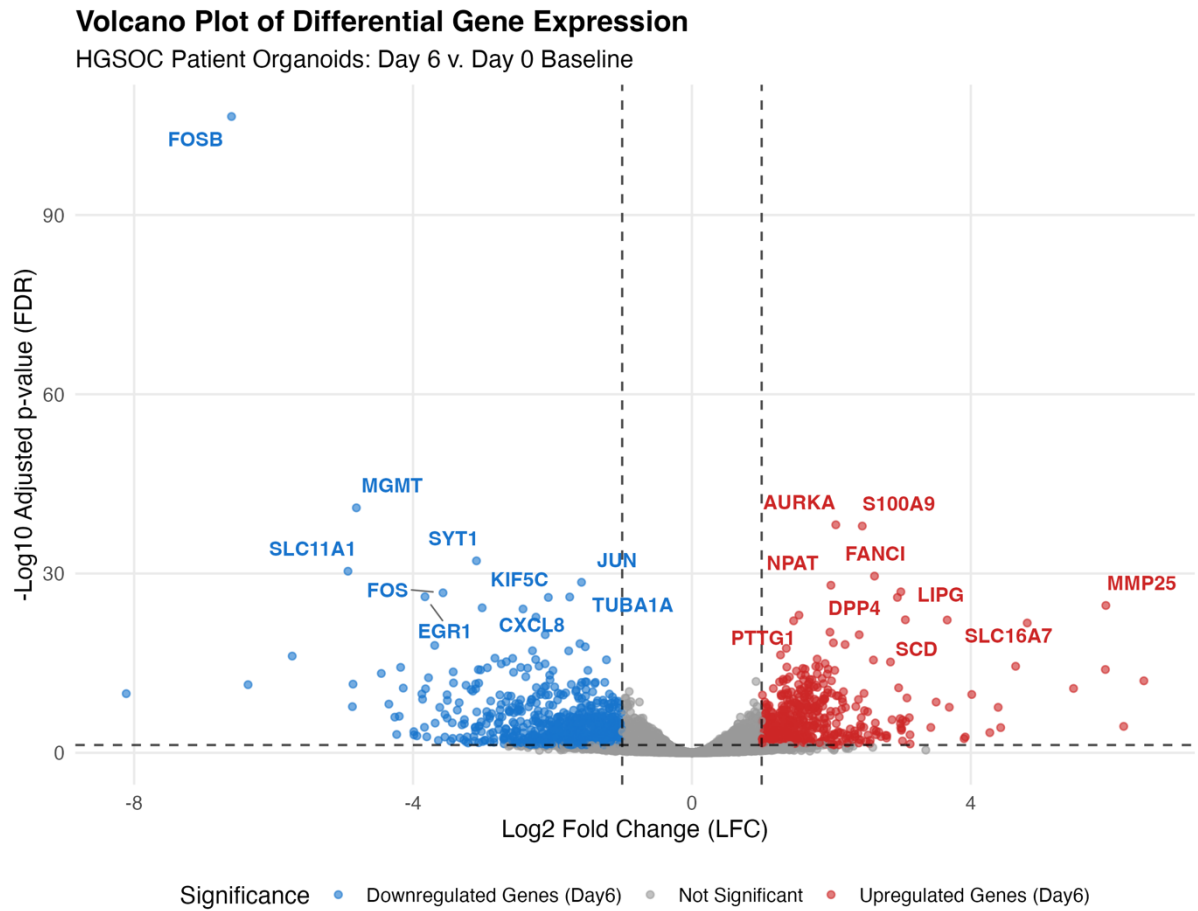


**Figure 1. Schematic workflow of a patient-derived 3D organoid pipeline for functional precision medicine in HGSOC.** Malignant cells are isolated from the

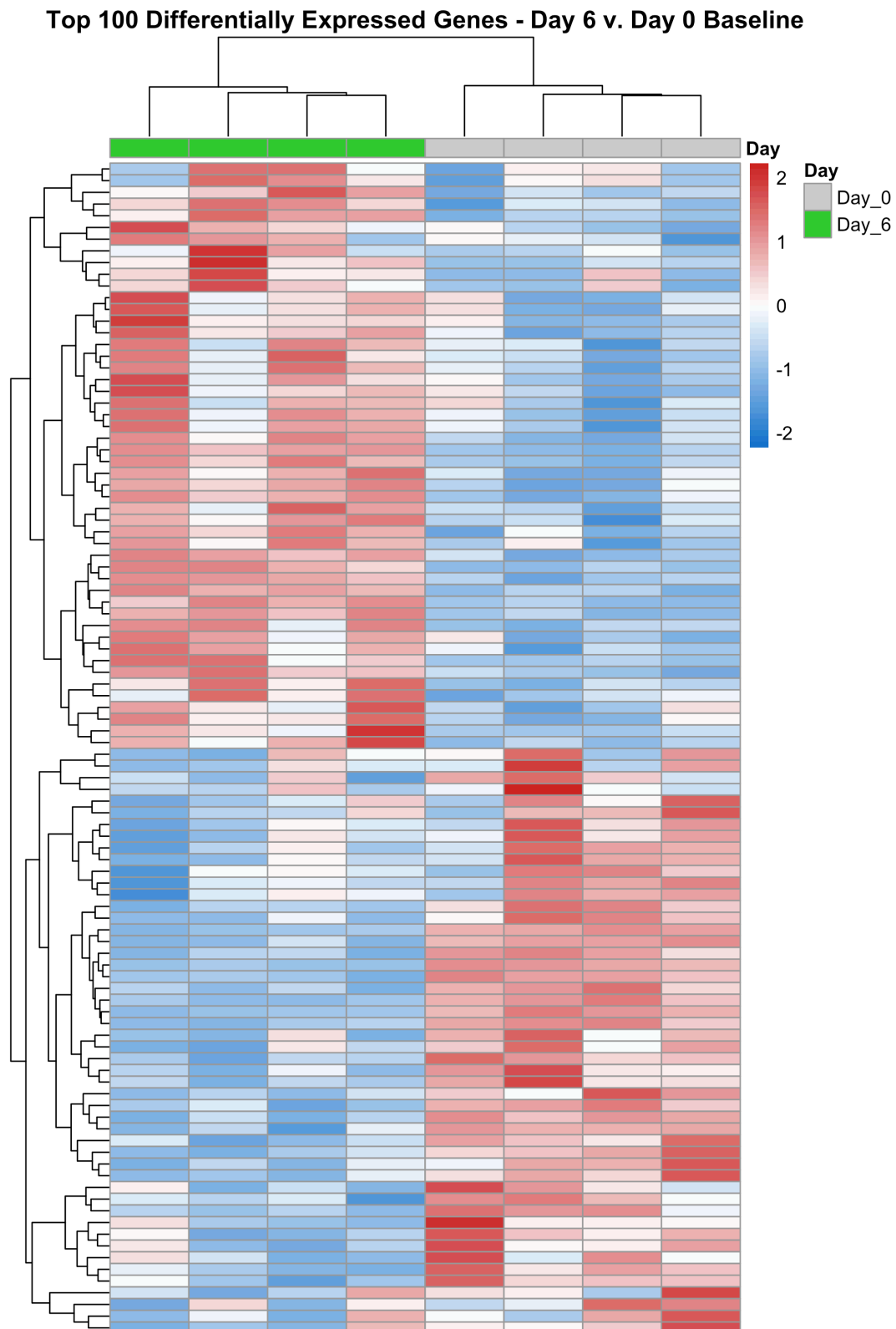
ascites or fluid surrounding the ovaries of a High-Grade Serous Ovarian Carcinoma (HGSOC) patient. The cells are cultured as 3D tumor organoids for 6 days in a nutrient-rich medium to establish structural and genomic architecture. Following initial culture, organoids undergo tissue dissociation and are replated into multi-well plates for seeding and expansion. Automated phenotypic quality control (QC) validates cell viability and biomarker expression prior to high-throughput drug library screening. The resulting therapeutic profile translates directly into personalized cancer treatments for the patient.



**Figure 5. Principal Component Analysis (PCA) of HGSOC Organoids.** PCA of variance-stabilized RNA-seq data showing global transcriptomic variation. Samples are coloured by patient and shaped by timepoint (Day 0 vs. Day 6). **Strong clustering by patient indicates that patient-specific differences are the primary source of variation.**

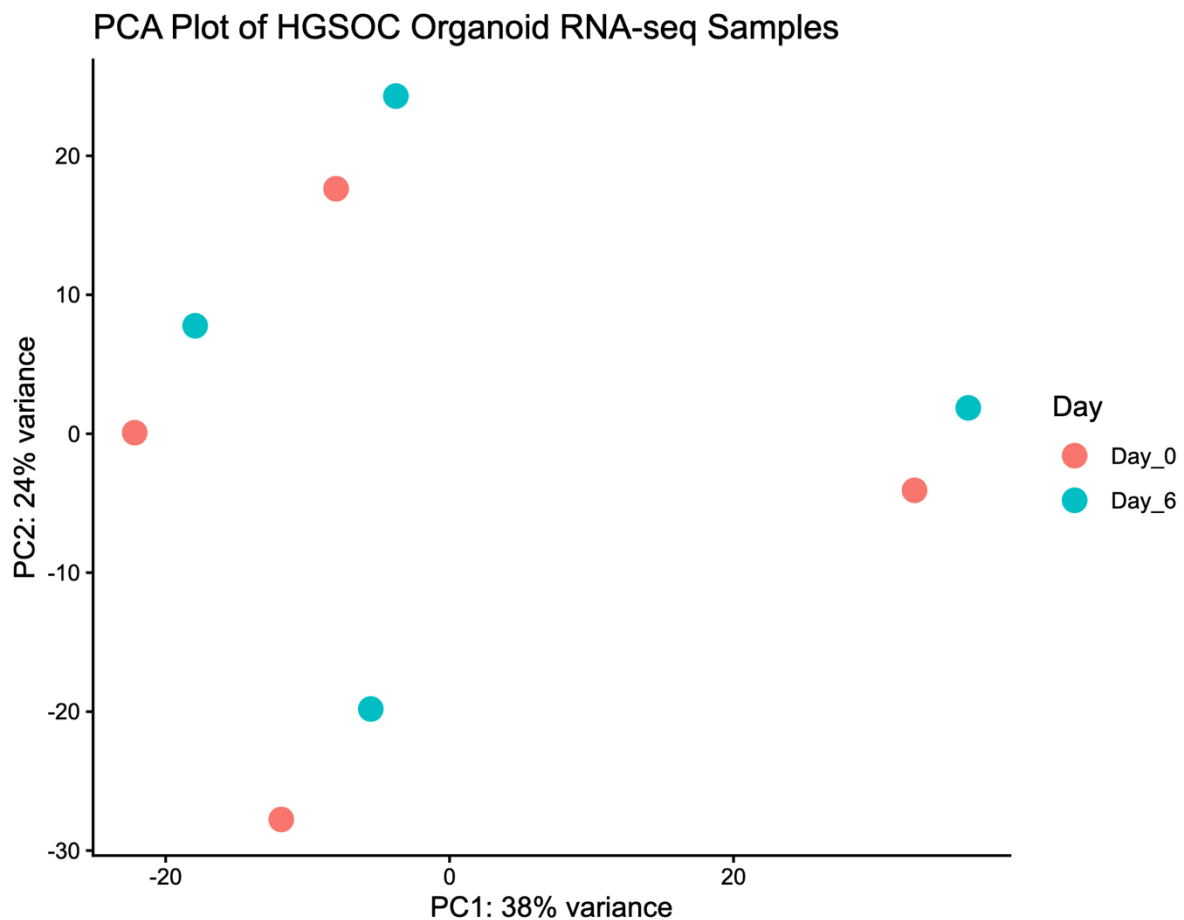


**Figure 6. Volcano Plot of Differential Gene Expression (Day 6 vs. Day 0 baseline) across all HGSOC patient-derived organoids.** Each point represents a gene, illustrating the overall transcriptomic changes associated with organoid culture over time.



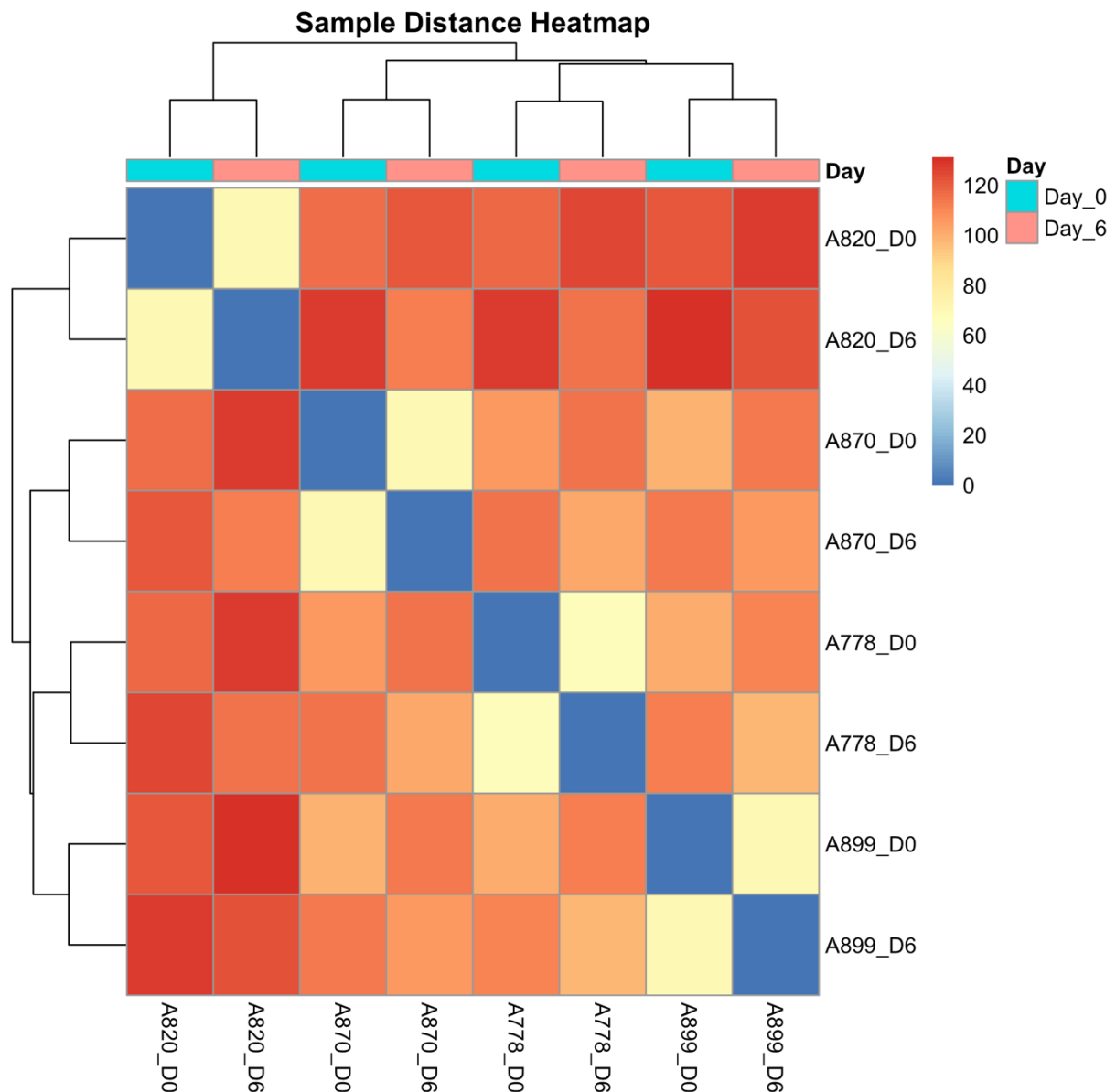
**Figure 7. Heatmap of the Top 100 Differentially Expressed Genes (Day 6 vs. Day 0).** Samples cluster perfectly by timepoint (top dendrogram). The genes split into two clear blocks: ~60% upregulated and ~40% downregulated at Day 6 (left dendrogram). Colours indicate relative expression variations (Z-scores). **Deep Red (+2)** = Upregulated, **White (0)** = Average Expression, **Deep Blue (-2)**: Downregulated.



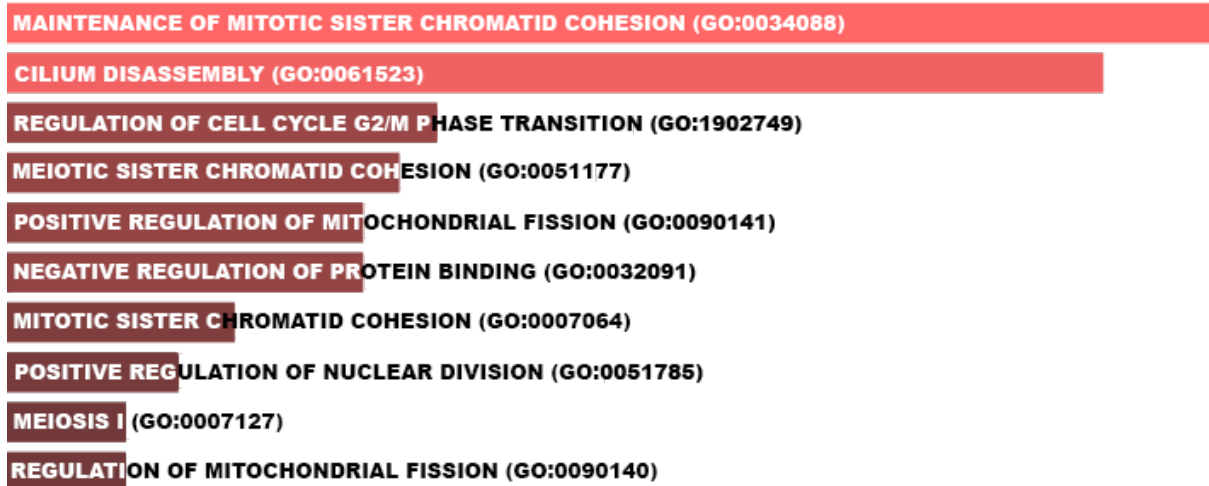


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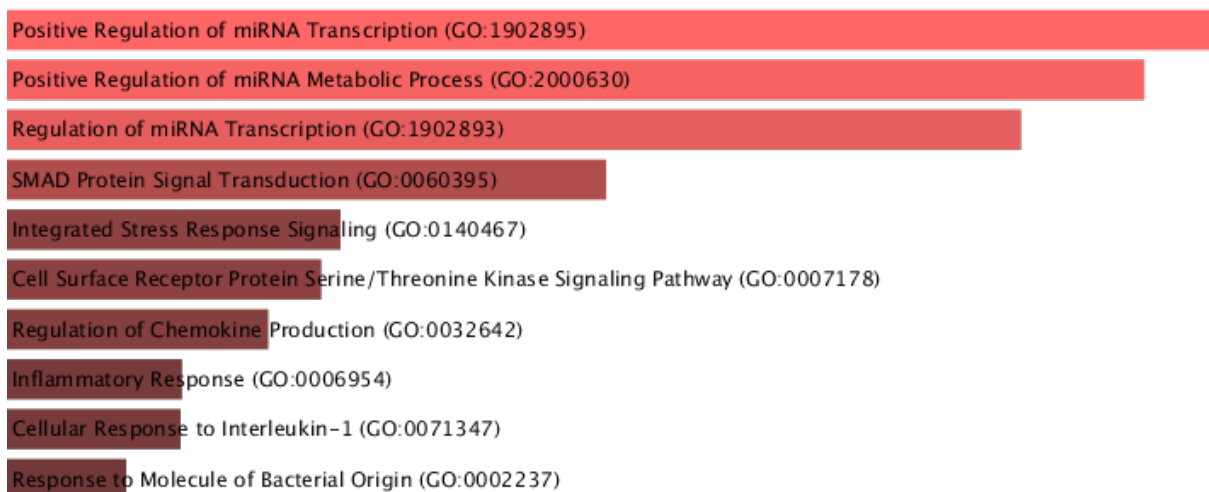




**Figure 9. Sample distance heatmap of HGSOc organoid transcriptomes across experimental conditions.** Hierarchical clustering analysis based on Euclidean distances computed from variance-stabilized RNA-Seq read counts across four patient-derived lines (A778, A820, A870, and A899). The colour gradient indicates relative distance, where darker blue signifies high transcriptional similarity (lower distance) and red signifies lower similarity (higher distance). The dendrogram demonstrates that samples cluster primarily by patient genetic background rather than by treatment timepoint (Day 0, cyan; Day 6, salmon). For each independent patient line, the Day 0 and Day 6 cultures cluster directly together as immediate sister pairs, highlighting that inter-patient biological variability outweighs the transcriptomic shifts induced by the 6-day *in vitro* culture microenvironment.



**Figure 10. Gene Ontology Enrichment Analysis of Upregulated Genes (Day 6).** Top upregulated biological processes, showing exit from G0 (cilium disassembly) and progression into active mitosis (chromosome cohesion, G2/M transition). Essentially, The cells exited quiescence and entered a phase of rapid proliferation.



**Figure 11. Gene Ontology Enrichment Analysis of Downregulated Genes (Day 6).** Enriched GO terms confirm the loss of host microenvironmental stress and immune signaling, characterized by significant reductions in integrated stress response signaling, inflammatory response, and chemokine production. This confirms the exit of the cells from an acute inflammatory and stressed state (characteristic of patient malignant effusion fluid).

## 14. Core Concepts

### 14.1. Biological & Disease related core concepts.

| Core Concept                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High-grade serous ovarian carcinoma (HGSOC)</b> | <p>High-grade serous ovarian carcinoma (HGSOC), the deadliest and most common form of ovarian cancer, is plagued by chemotherapy resistance, making physiologically relevant models crucial for clinical research(Tadić et al., 2024).</p> <p>Globally, ~ 600,000 women live with HGSOC.<sup>2</sup> (Ren et al. 2025).</p>                                                                                                                                                                                                                                                                                          |
| <b>Patient-derived 3D organoids (PDOs)</b>         | <p>An organoid is a tiny, 3D "mini-tumor" or "mini-organ" grown in a dish from a patient's actual cells.</p> <p>Historically, if researched wanted to test a cancer drug in a lab, they grew tumor cells flat on a plastic dish. However, cells don't live on flat plastic inside our bodies; they live in a complex, 3D world. Flat cells behave differently and don't preserve the true patient tumor architecture and biology, hence are not useful in drug screening.</p> <p>PDOs are premier tools and vastly outperform traditional 2D cultures by preserving true patient tumor architecture and biology.</p> |
| <b><i>In vivo</i></b>                              | <p>Experiments performed in or on a living organism (e.g., a human, animal or plant), as opposed to in a petri dish or test tube, where conditions are highly controlled and isolated. <i>In vivo</i> experiments give us insight into biological context.</p>                                                                                                                                                                                                                                                                                                                                                       |
| <b><i>In vitro</i></b>                             | <p>Experiments performed outside of a living organism, such as in a petri dish or test tube, where conditions are highly controlled and isolated. <i>In vitro</i> experiments may not necessarily provide as much biological insight as <i>in vivo</i> experiments.</p>                                                                                                                                                                                                                                                                                                                                              |
| <b>Tumor Microenvironment (TME)</b>                | <p>The tumor microenvironment (TME) is the local cellular and biochemical environment surrounding a tumor inside the patient's body.</p> <p>In HGSOC, the native TME encompasses the nutrient-stressed and hypoxic malignant effusion fluid, stromal fibroblasts, immune cells, endothelial cells, extracellular matrix, and various signaling cytokines.</p> <p>Transitioning cells out of this complex TME (<i>in vivo</i> conditions) to <i>in vitro</i> conditions is what triggers metabolic reprogramming.</p>                                                                                                 |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Metabolic reprogramming</b>      | <p>The process by which cellular metabolism is redirected to support proliferation rather than stress adaptation. This occurs when cells transition from <i>in vivo</i> tumor microenvironment (characterized by low oxygen (hypoxia), nutrient gradients, stromal fibroblasts, immune cells, endothelial cells, cytokines and growth factors) to an <i>in vitro</i>, nutrient-rich environment, where culture media contains glucose, amino acids, vitamins, and growth factors that promote growth.</p> <p>Once the cells adapt, they no longer need to devote as many resources to acute stress responses. Instead, they begin investing energy into DNA replication, biomass production and cell division</p> |
| <b>Hypoxia</b>                      | <p>A state of low oxygen conditions that is highly characteristic of the TME and malignant effusions <i>in vivo</i>. This low oxygen state forces cancer cells to adapt by activating cellular survival, anti-apoptotic, and inflammatory signaling pathways.</p> <p>Once cells transition to nutrient- and oxygen-rich <i>in vitro</i> culture, this hypoxic condition rapidly relieved, and associated genes are downregulated.</p>                                                                                                                                                                                                                                                                             |
| <b>Immediate Early Genes (IEGs)</b> | <p>These genes represent a class of transiently and rapidly activated genes (such as <i>FOS</i>, <i>FOSB</i>, <i>JUN</i>, and <i>EGR1</i>) that drive the cell's primary genomic response to environmental change, physical disruption, or mechanical isolation stress.</p> <p>These early-response genes spike during tissue harvesting (Day 0) and level off once HGSOc cells stabilize in culture (Day 6).</p> <p>Therefore, they serve as key transcriptomic markers of microenvironmental adaptation.</p>                                                                                                                                                                                                    |
| <b>Malignant Effusion (Ascites)</b> | <p>The protein- and cell-rich fluid in the peritoneal cavity of HGSOc patients that serves as the native, nutrient-stressed, and hypoxic <i>in vivo</i> environment from which primary HGSOc cells are isolated before being transitioned to <i>in vitro</i> 3D organoid cultures.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Precision Oncology</b>           | <p>The premise of precision oncology (and precision medicine, in general), is moving from a one-size-fits-all model of treatment toward a personalized form of treatment focused on a patient's unique genetic, transcriptomic, and molecular profile.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Translational Medicine</b>       | <p>The practice of translational medicine focuses on translating the findings of laboratory experiments into real-world clinical applications (i.e., medical treatments, diagnostics, and interventions).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|  | <p>Patient-derived organoids (PDOs) are the ultimate functional bridge between translational medicine and precision oncology. By cultivating a patient's own tumor cells into 3D organoids, researchers can preserve the exact genetic, transcriptomic, and molecular heterogeneity of the patient's unique disease.</p> <p>In the context of translational medicine, these 3D models allow researchers to bridge the gap from drug discoveries in the lab to administering them to actual patients (after clinical trials and rigorous regulatory approval), generating insightful and biologically-relevant data.</p> <p>This allows clinicians to screen hundreds of therapeutic targets <i>in vitro</i> to predict how a patient's actual tumor will respond to treatment in theory, before in practice.</p> |
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14.2. Statistical & Bioinformatics related core concepts.

| Core Concept                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p><b>RNA-sequencing (RNA-seq)</b></p>     | <p>At its core, RNA-sequencing (RNA-seq) is a high-throughput technology used to detect, analyze, and quantify the entire complement of RNA transcripts in a biological sample (the transcriptome) at a given moment.</p> <p>By converting RNA into stable complementary DNA (cDNA) libraries and sequencing them, it lets us look past the static blueprint of the genome to see exactly which genes are actively turned on, turned off, or spliced differently under specific conditions.</p> <p>At its core, RNA-sequencing (RNA-seq) is a high-throughput technology designed to profile and quantify the entire transcriptome of a biological sample at a specific point in time. By converting unstable RNA into stable complementary DNA (cDNA) libraries for sequencing, this method moves beyond the static genome to reveal precise, real-time dynamics, specifically, which genes are actively transcribed, silenced, or alternatively spliced under distinct conditions.</p> |
| <p><b>Differential Gene Expression</b></p> | <p>DGEA is used to determine which genes show statistically significant changes in expression levels between different experimental conditions (for example, comparing the HGSOC organoids at Day 0 versus Day 6).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <p><b>Generalized Linear Model (GLM)</b></p> | <p>The overarching statistical framework used to predict gene expression changes based on the experimental design.</p> <p>While standard linear regression assumes the data follows a bell curve (normal distribution), a GLM provides the flexibility to accommodate non-normal data, specifically, the Negative Binomial distribution of the raw read counts.</p> <p>Ultimately, GLM allows DESeq2 to map experimental variables directly onto the Negative Binomial distribution of raw sequencing counts.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Paired Experimental Design</b></p>     | <p>An experimental design that involves comparing samples from the exact same patient line across different timepoints, specifically matching each Day 6 sample back to its own Day 0 baseline.</p> <p>Human genetic diversity is massive. Every human being has a completely unique genetic blueprint. The baseline differences between Patient A and Patient B are vastly larger than the subtle transcriptomic changes that occur between Day 0 and Day 6. Running a standard, unpaired analysis would cause the huge genetic "noise" of patient-to-patient variation to completely drown out the "signal" of their 6-day adaptation.</p> <p>Therefore, by using a "paired" statistical approach, which essentially compares each patient's lab cells <i>only</i> back to their own original cells, the background noise is muted. This allows researchers pinpoint the exact genes that change when cells adjust to lab conditions.</p> <p>This approach significantly increases statistical power by using patients as their own baselines.</p> |
| <p><b>Shapiro-Wilks Test</b></p>             | <p>A statistical test used to determine if a dataset is normally distributed, by evaluating a single numeric variable against two hypotheses:</p> <ul style="list-style-type: none"> <li>• <b>Null Hypothesis (<math>H_0</math>):</b> The data is normally distributed.</li> <li>• <b>Alternative Hypothesis (<math>H_a</math>):</b> The data is not normally distributed.</li> </ul> <p><b>Interpretation of the Shapiro-Wilk Test:</b></p> <ul style="list-style-type: none"> <li>• If the <math>p &gt; 0.05</math> = Failure to reject null hypothesis. This means that the data is roughly normally distributed, and therefore, it is safe to use parametric tests like t-tests or ANOVA.</li> </ul>                                                                                                                                                                                                                                                                                                                                             |

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|                                                                 | <ul style="list-style-type: none"> <li><math>p \leq 0.05</math> = Reject the null hypothesis. This means that the data significantly deviates from a normal distribution, and non-parametric alternatives like Wilcoxon or Mann-Whitney may be required.</li> </ul>                                                                                                                                                                                                                                                                                                                       |
| <b>Wald Test</b>                                                | <p>A statistical method used to evaluate whether specific parameters in a model (such as regression coefficients) are significantly different from zero or a hypothesized value. Essentially, it determines whether removing or restricting those parameters would meaningfully degrade the model's fit.</p> <p>The Wald test produces a raw p-value for each gene, which is then adjusted/corrected by the Benjamini-Hochberg (BH) FDR correction to control for false positives. Significantly differentially expressed genes are determined after this BH FDR correction.</p>          |
| <b>Benjamini–Hochberg False Discovery Rate (FDR) Correction</b> | <p>The Benjamini–Hochberg False Discovery Rate (FDR) correction is a statistical method used to control the inflation of false positives, when analyzing thousands of genes, simultaneously. Without correction, unadjusted significance threshold of <math>p &lt; 0.05</math> would yield thousands of false positives by random chance alone.</p> <p>The BH procedure addresses this issue by adjusting p-values when performing many hypothesis tests simultaneously, such as in this analysis, where we are performing differential expression analysis on thousands of genes.</p>    |
| <b>Variance Stabilizing Transformation/Testing (VST)</b>        | <p>A normalization transformation method used to adjust gene expression data so that very highly expressed genes do not overpower the analysis. This gives all genes a more equal chance to contribute, making patterns between samples easier to identify.</p> <p>Simply, VST ensures that the variance is not dominated by high-expression genes, and that the data is homoscedastic (i.e., has a constant variance across the entire range of mean values). This prevents highly expressed genes from dominating downstream analyses (in this case the quality control PCA below).</p> |
| <b>Log Transformation</b>                                       | <p>Raw RNA-seq data is highly skewed. Plotting these raw sequencing read counts on the same graph will result in the highly expressed genes will “shoot” off the graph, while lowly-expressed genes will get “squished” into a virtually invisible line at the bottom of the graph.</p> <p>Log-transformation (specifically <math>\log_2</math> transformation) is a crucial step used to convert this highly-skewed, exponential gene expression data into manageable and</p>                                                                                                            |



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|                      | <p>statistically stable data. Log transformation essentially compresses the highly expressed genes, and “stretches” the lowly expressed genes, such that the graph becomes more symmetrical.</p> <p>This symmetry allows for easier plotting, interpretation, and visual comparison of up-regulated genes (such as <i>AURKA</i>) against down-regulated genes (such as <i>JUN</i>).</p> <p>Additionally, log transformation stabilizes variance (homoscedasticity), ensuring that highly expressed genes don't dominate the downstream statistical analysis purely because of their scale.</p>                                                                                                                         |
| <b>Normality</b>     | <p>This mean that the data follows a normal (i.e., Gaussian) distribution, characterized by the symmetric “bell” curve, with the peak representing the mean of the data, where most of the data clusters. Outliers taper off equally in both directions.</p> <p>Histograms can be used to visually inspect the normality of a dataset. If the shape is roughly bell-shaped, the data is normal.</p> <p>The Shapiro-Wilk test is a statistical test used to determine if a dataset is normally distributed. If the <math>p &gt; 0.05</math> = Failure to reject null hypothesis. This means that the data is roughly normally distributed, and therefore, it is safe to use parametric tests like t-tests or ANOVA.</p> |
| <b>Normalization</b> | <p>A statistical method of bringing all of our data to a comparable scale, particularly in cases where the features on our dataset have a wide range, for example, a dataset looking at daily cell count in an in vitro organoid culture (ranging from 1,000 to 100,000 cells).</p> <p>This is an issue for downstream analysis tools like KNN that rely on distance calculation.</p>                                                                                                                                                                                                                                                                                                                                  |
| <b>p-value</b>       | <p>As it relates to RNA-seq data and statistical tests, the p-value is a probability between 0 and 1 that measures the strength of our experimental results against the null hypothesis (i.e., baseline assumption).</p> <p><b>If:</b></p> <ul style="list-style-type: none"> <li>• <math>p &gt; 0.05</math> = Failure to reject null hypothesis (e.g., the gene's expression change is NOT statistically significant, and any variation could be due to “background noise”.</li> <li>• <math>p \leq 0.05</math> = Reject the null hypothesis (e.g., the gene's expression change is statistically significant).</li> </ul>                                                                                            |



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| <p><b>P Adjusted Value (<math>p_{adj}</math>)</b></p> | <p>Also known as FDR (False Discovery Rate) adjusts the raw p-values to control for false discoveries (i.e., false positives) across a given dataset.</p> <p>A standard <math>p &lt; 0.05</math> carries a 5% false-positive risk. Testing 20,000 genes yields roughly 1,000 false positives purely by chance.</p> <p>Adjusted p-values (<math>p_{adj}</math>) correct for this multiple testing problem by penalizing raw values, ensuring no more than 5% of the significant genes are false discoveries.</p>                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Z-Score</b></p>                                 | <p>The Z-score (standard score) describes a given data point's relationship to the mean of a group of values. It tells you how many standard deviations a single data point lies above or below the mean. The Z-score is a crucial tool for normalizing data, as it allows us to compare scores from entirely different datasets or distributions on a level playing field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Principal Component Analysis (PCA)</b></p>      | <p>A statistical technique used to reduce the number of features (i.e., dimensionality reduction) and visualize high-dimensional data in 2D or 3D. It does so by transforming a high-dimensional dataset with many correlated variables into a smaller set of uncorrelated variables called principal components, while reducing noise, removing multicollinearity, and preserving as much of the original data's variability as possible.</p> <ul style="list-style-type: none"> <li>• <b>First Principal Component (PC1) (primary axis)</b> captures the maximum possible variance (or most dominant pattern) in the data.</li> <li>• <b>Second Principal Component (PC2):</b> Captures the next highest variance and is perpendicular to PC1, ensuring that the information it captures is completely independent and does not overlap with the first component.</li> </ul> |
| <p><b>Volcano Plot</b></p>                            | <p>A volcano plot allows visualization of the results of a differential gene expression analysis.</p> <p>The “volcano” shape is due to the fact that highly significant genes with large fold changes are plotted in the upper-left and upper-right corners of the plot, forming a distinct “V” shape that resembles an erupting volcano.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Heatmap</b></p>                                 | <p>A heatmap also allows visualization of the results of a differential gene expression analysis, by showing how genes cluster across individual samples or replicates.</p> <p>The dendrograms on the top and left sides of the plot show which genes (rows) cluster together. Individual cells in the matrix are colour-coded by changes in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                       | relative expression. This colour-coding allows for easy visualization of patterns in our dataset.                                                                                                                                                                                                                                                                                               |
| <b>Dendrogram</b>                     | <p>A dendrogram is a tree-like branching diagram used in hierarchical clustering to visualizes sample and gene relationships on the heatmap, and essentially illustrate relationships between clusters.</p> <p>The length of the branches represents the degree of transcriptomic similarity or mathematical distance between genes or samples, allowing researchers to visualize patterns.</p> |
| <b>Negative Binomial Distribution</b> | <p>Raw RNA-seq read counts are highly skewed, non-negative, and show a strong mean-variance relationship. As a result standard normal distributions fail.</p> <p>DESeq2 uses the Negative Binomial distribution (a discrete probability distribution) to mathematically model the over-dispersed count data, where the variance is larger than the mean.</p>                                    |